Spatial diffusion of innovation – agents of change, diversification networks, spacial life cycles

Economic Complexity, Class 8 – Nov 11 2026

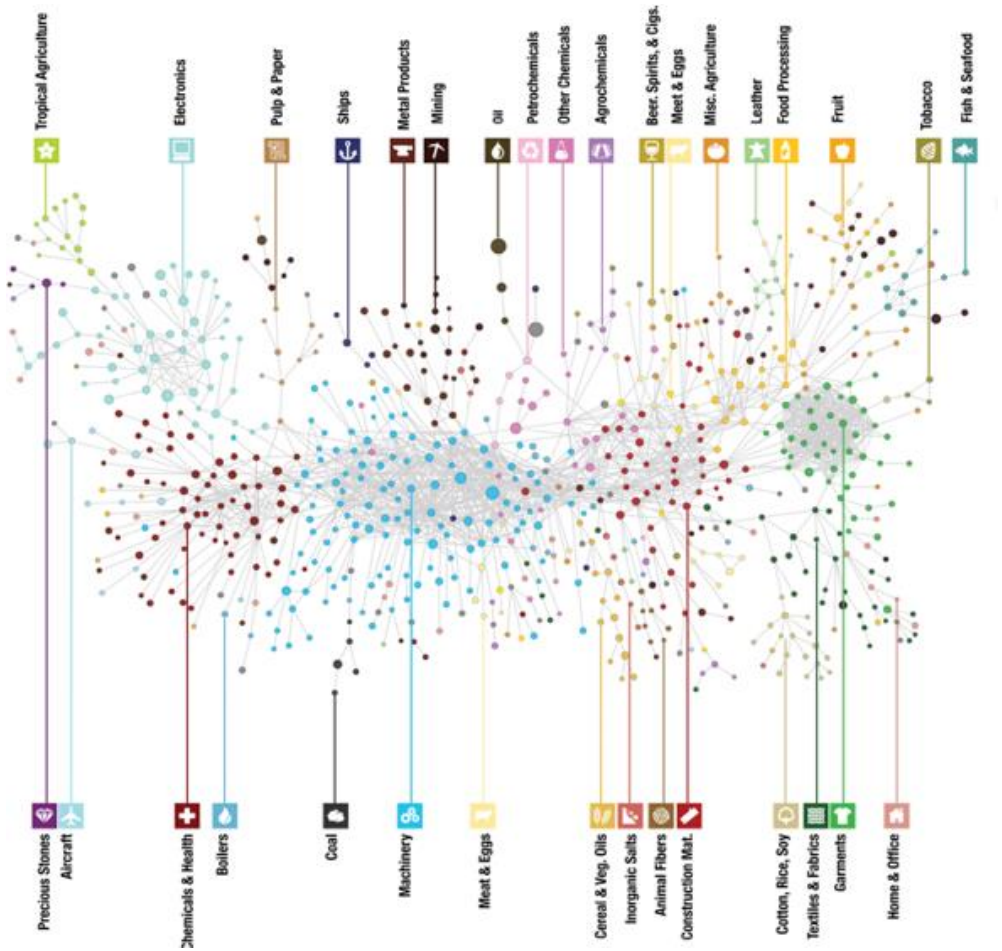
Social Data Science, Corvinus University of Budapest

Today

1. Agents of change
2. Collaboration networks in diversification
3. Life cycles

1. Agents of change

How do we jump big in the product/technology/knowledge space?

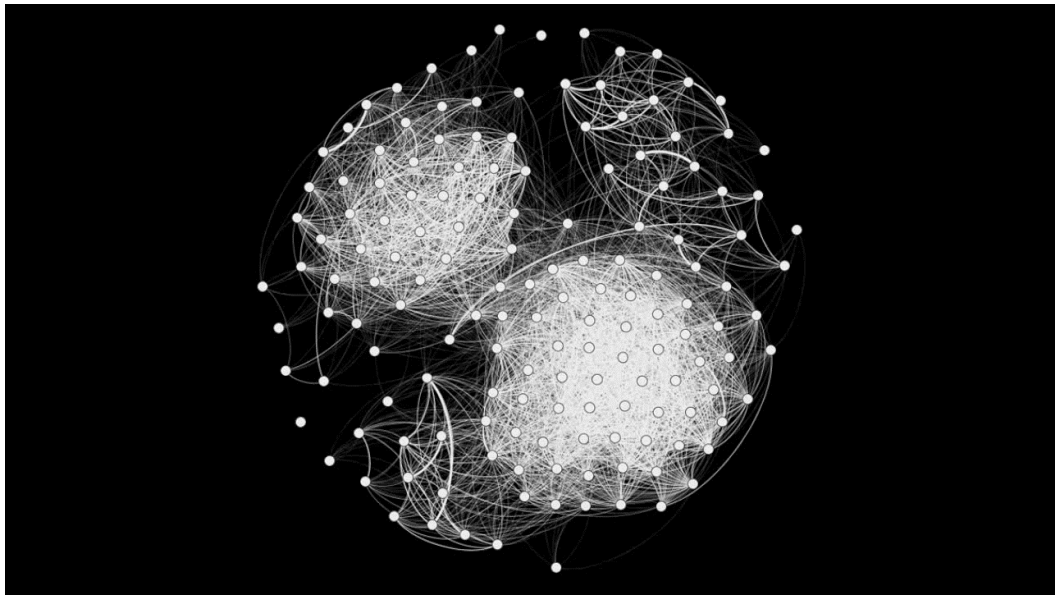


Hidalgo et al. Science (2007)

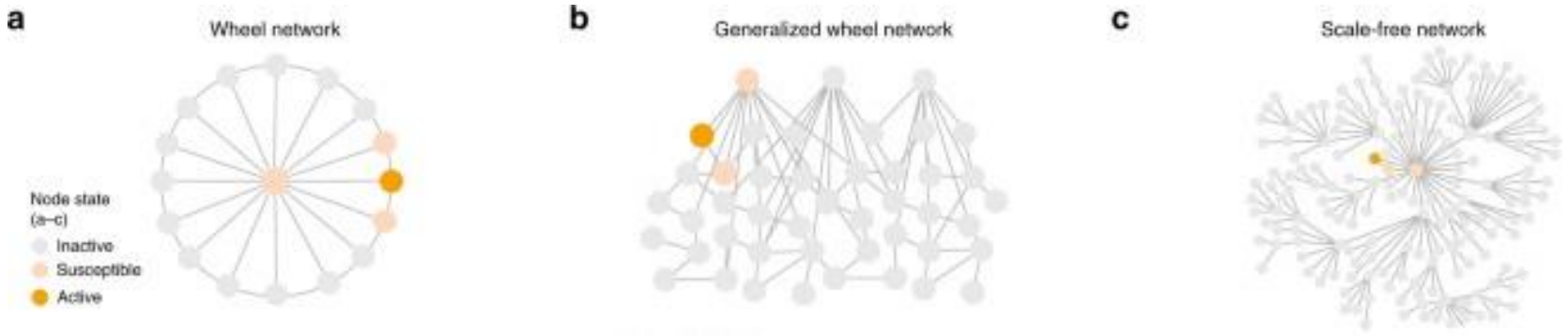
Unrelated diversification

- Investments?
  - FDI
  - Governments
- Migration?
  - Inventors
  - Scholars
  - Businessmen

Diversification as spatial diffusion?



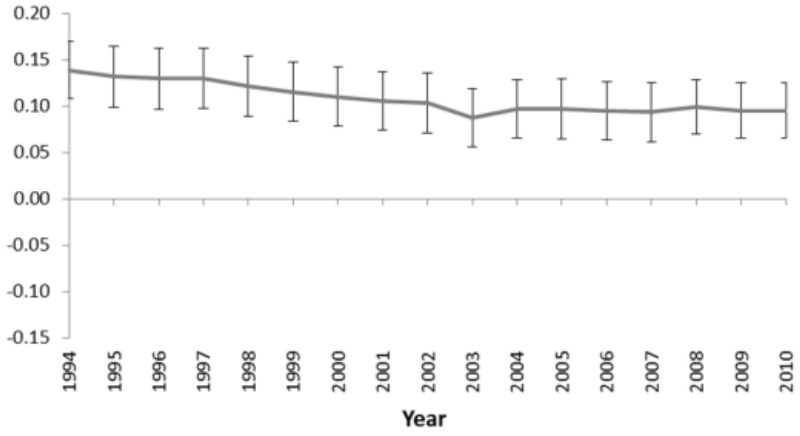
Aplin et al. (2015) Nature



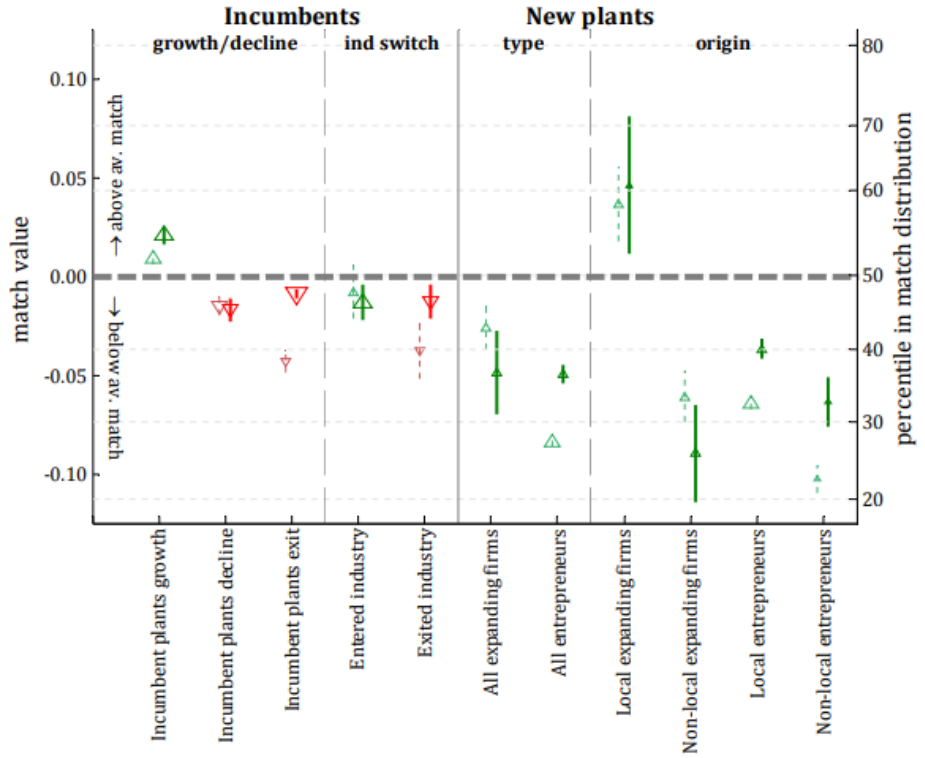
Alshamsi, Pinhero, Hidalgo (2018) Nat. Comm.

Structural change by type of firms

Figure 5: Structural change in Sweden's labor market regions



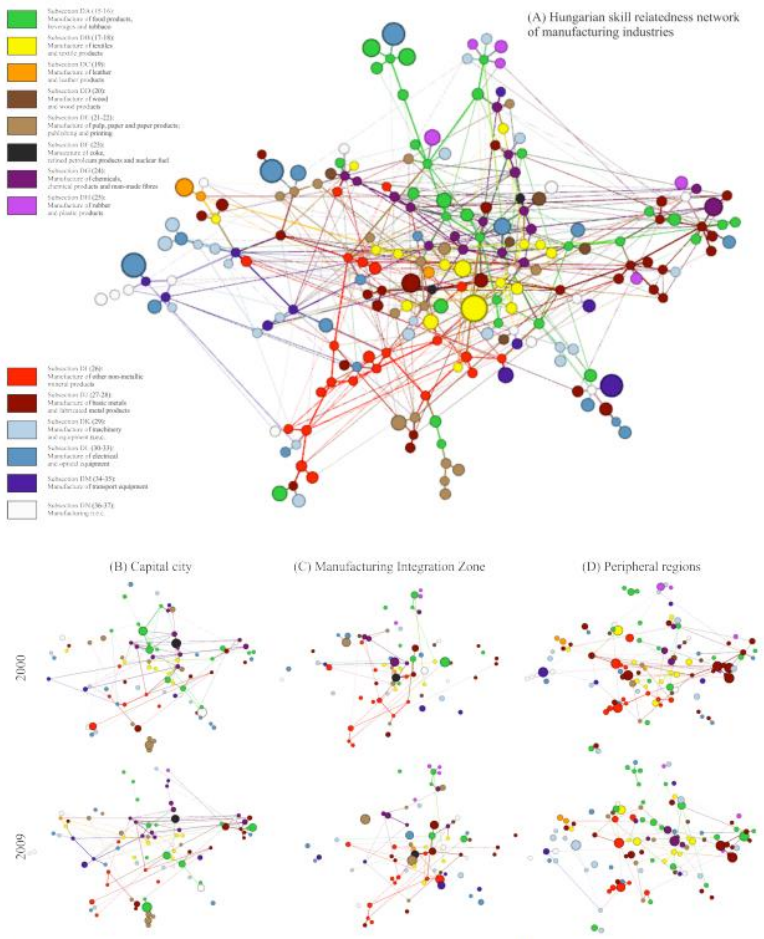
The graph depicts the development of the average capability match of a regions' local industries to the local economy of the base year 1994 ($S_{rt,1994} = \sum_i \frac{E_{irt}}{E_{rt}} \widetilde{LQ}^{rel}_{ir1994}$), including 95% error bars. In the presence of structural change, $S_{rt,1994}$ should fall (diversification) or rise (increasing of existing focus) over time. The error bars depict a 95% confidence interval calculated as ±1.96 times the standard deviation of the mean structural change across regions



The markers shows for each agent type the employment weighted average capability match of the local industries in which the agent type creates or destroys employment within ten years after being founded, averaged across all establishments of the agent minus the region's coherence. That is, the graph shows by how much an agent's average capability match exceeds the region's coherence. As a consequence, employment created (destroyed) at $\tilde{A} > 0$ corresponds to a diversification (further focusing) of the regional capability base. Employment-creating agents are denoted with a green, upward pointing arrow, employment-destroying agents with a red downward-pointing arrow. The error bars depict a 95% confidence interval, based on the standard deviation of the mean capability match across all establishments of a given agent type. To facilitate interpretation of $\tilde{A}$-values, the right most vertical axis provides the corresponding percentiles of a capability match value in the overall distribution of the 1994 economy. The one-year figures reported in Figure 6 are provided as a reference with lighter markers and dotted confidence intervals.

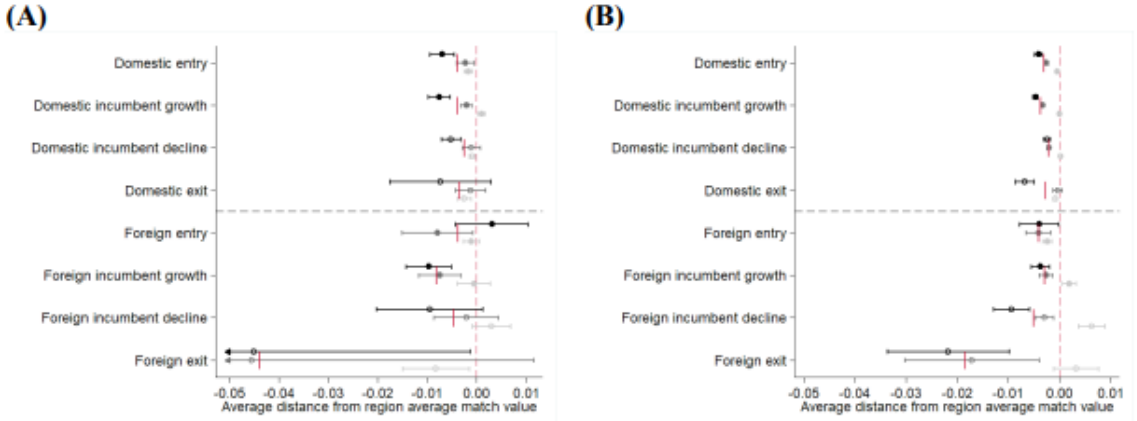
Neffke et al. 2017 Economic Geography

Foreign firms and structural change



Notes: **(A)** The network representation was constructed by superimposing the strongest 1.5% of edges on the maximum spanning tree of the skill relatedness network. ForceAtlas2 layout in Gephi software was used as layout. After obtaining the network layout, additional ties up to the strongest 5% of all edges were added to the graph. Node size is proportional to the number of employees in each industry. **(B-D)** Node size represents employment location quotient; only those industries are represented that have $LQ > 1$.

Figure 6. Structural change by agent type in the 3 region groups



Notes: **(A)** Short-term (1-year) structural change. **(B)** Long-term (10-year) structural change. The red dashed line indicates the average distance of agent types from the region average match value. Values to the right of this line indicate more related (*i.e.* above average) diversification, while values to the left indicate more unrelated (*i.e.* below average) diversification. Error bars indicate 95% confidence intervals. The base year is 2000. Black error bars depict the *Manufacturing Integration Zone*, dark grey error bars indicate the *peripheral regions*, light grey bars signify the *capital city*. Red solid vertical lines indicate the agent type-average structural change.

Elekes et al. 2019 Regional Studies

2. Networks


BASED ON THE BEST-SELLING NOVEL BY THE AUTHOR OF
THE DA VINCI CODE AND ANGELS & DEMONS
TOM HANKS
FELICITY JONES
INFERNO
IS UPON US


JENNIFER LAWRENCE
JOEL EDGERTON
MATTHIAS SCHOENAERTS
JEREMY IRONS
RED SPARROW
MARCH 2


IT BEGINS
DUNE
IN THEATERS AND IMAX
OCTOBER 22


THEY SAY NO ONE CAN SAVE THE WORLD.
MEET NO ONE.
RYAN REYNOLDS
A MICHAEL BAY FILM
6 UNDERGROUND
DECEMBER 13 | NETFLIX


WILL SMITH
WILL SMITH
WHO WILL SAVE YOU FROM YOURSELF
GEMINI MAN
IN IMAX, 3D & 2D CINEMAS
COMING SOON


RYAN GOSLING
HARRISON FORD
BLADE RUNNER 2049
10.6.17


THE KING
ALL HAIL
IN SELECT THEATERS AND ON
NETFLIX | THIS FALL


YOU DON'T NEED A PAST
TO HAVE A FUTURE
VIN DIESEL
BLOODSHOT
BASED ON THE BEST-SELLING COMIC BOOK
SUPERHEROES ARE GETTING AN UPGRADE
MARCH 13


WELCOME TO THE DAY AFTER JUDGMENT DAY
TERMINATOR
DARK FATE


THE WITCHER

Local capabilities and interregional connections









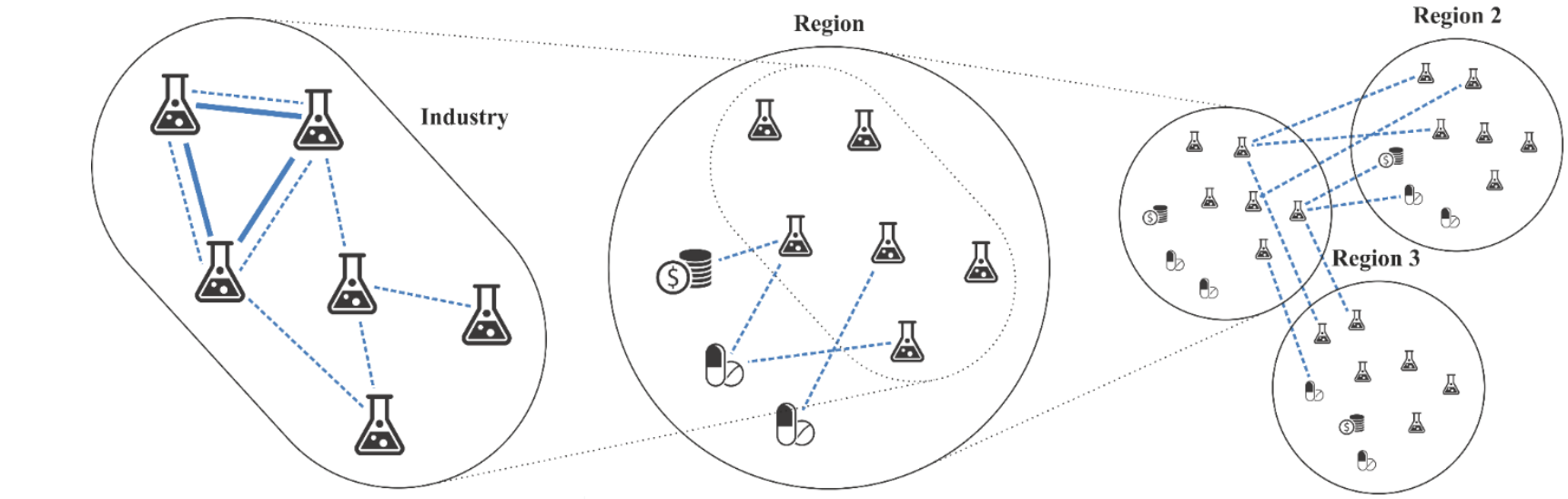


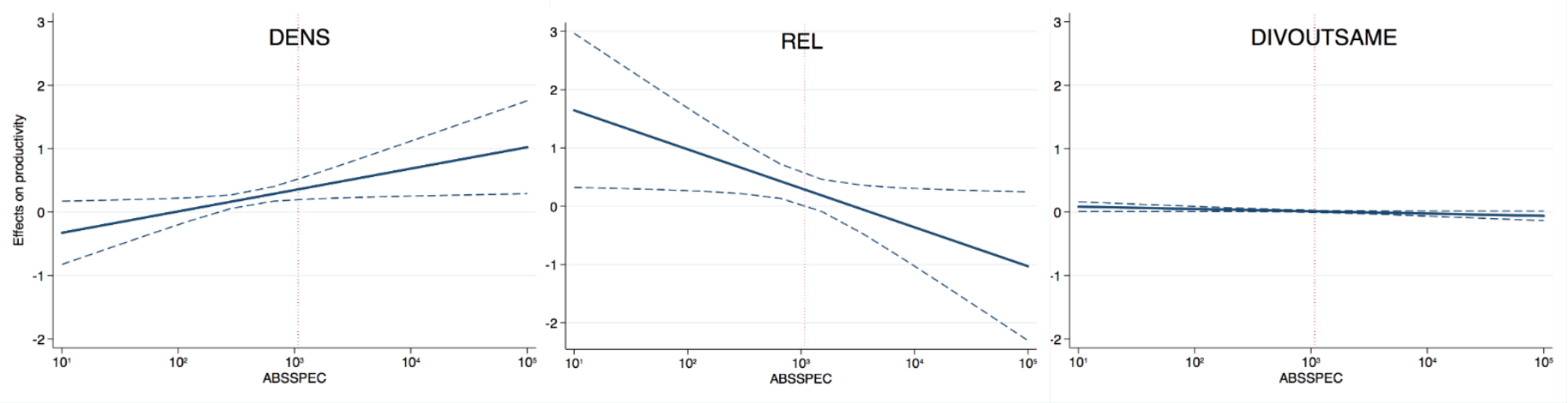




Knowledge sourcing in diversification

- The effect of co-worker networks on regional growth of income depends on region size
- (ABSSPEC)





Eriksson and Lengyel (2019) Economic Geography

Question 1

Can interregional networks

substitute

local capabilities in diversification?



Lőrincz et al. (2020) Applied Network Science

Firms as drivers of diversification

Agents of structural change: mobile entrepreneurs (Neffke et al., 2018); foreign-owned firms (Elekes et al., 2019), R&D investments (Crescenzi et al., 2022).

Interregional knowledge sourcing dominantly happens within boundaries of multilocation companies (Frigon and Rigby, 2022; Zhang and Rigby, 2022).

Multilocation companies intensify collaboration to source from diversity (Alcacer and Zhao, 2012).

Firms as drivers of diversification

Exploitation of the local knowledge base favors related diversification (Wang and Zhao, 2018).

Agents of structural change and co-worker networks literature suggest that regions can source unrelated knowledge from the diversity of regions that multilocation companies connect (Neffke et al., 2018; Elekes et al., 2019; Eriksson and Lengyel, 2019; Crescenzi et al., 2022).

Question 2

Do internal connections

of multilocation firms across regions

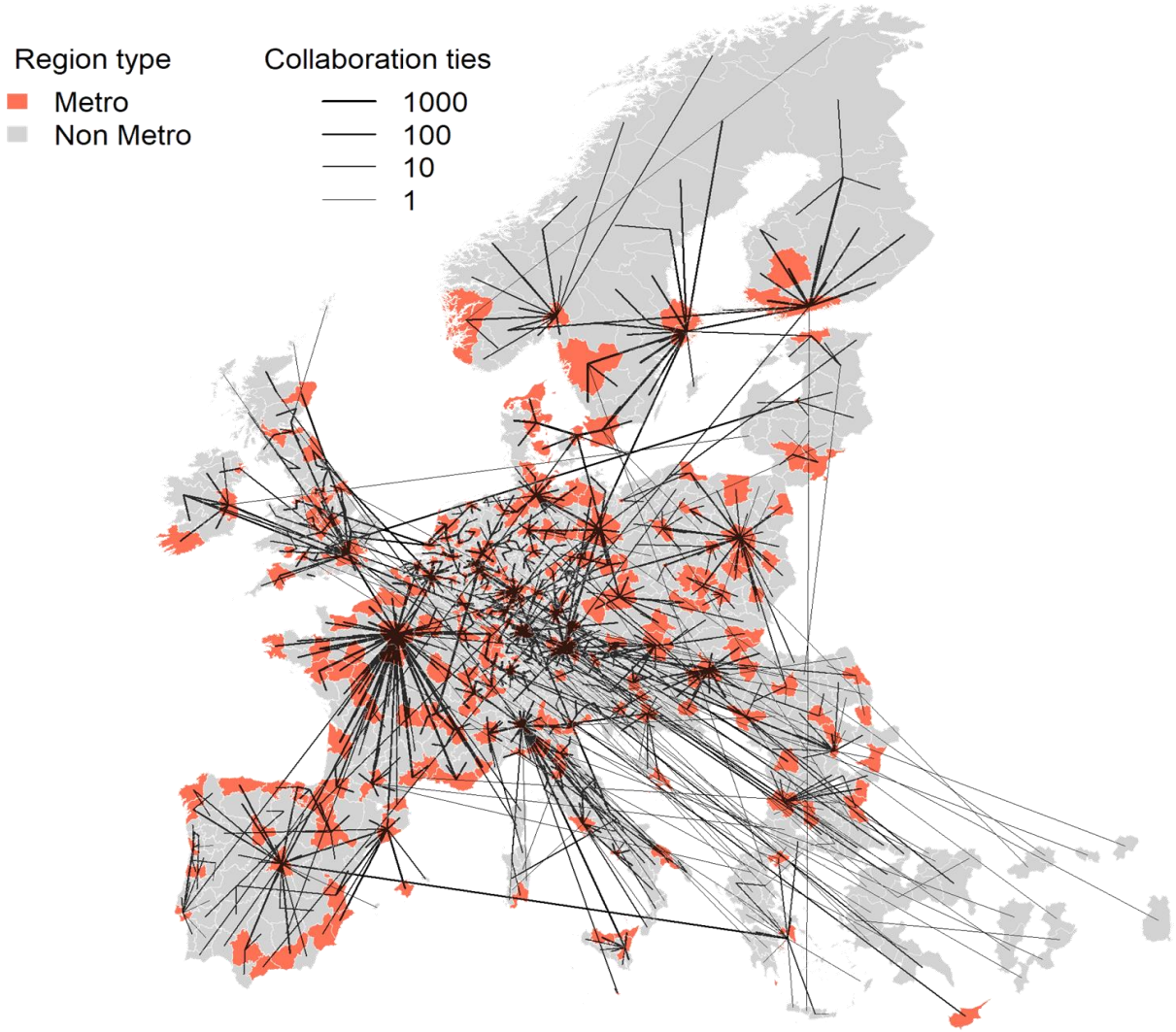
favors diversification

into related or unrelated new activities?



Lőrincz et al. (2020) Applied Network Science

Data



- EPO PATSTAT
- 1981-2015, five-year periods
- 4-digit CPC codes
- NUTS3 regions (EUROSTAT classificaiton of metro/nonmetro)
- Fractionally split patents to regions by inventor addresses

Probability of ENTRY

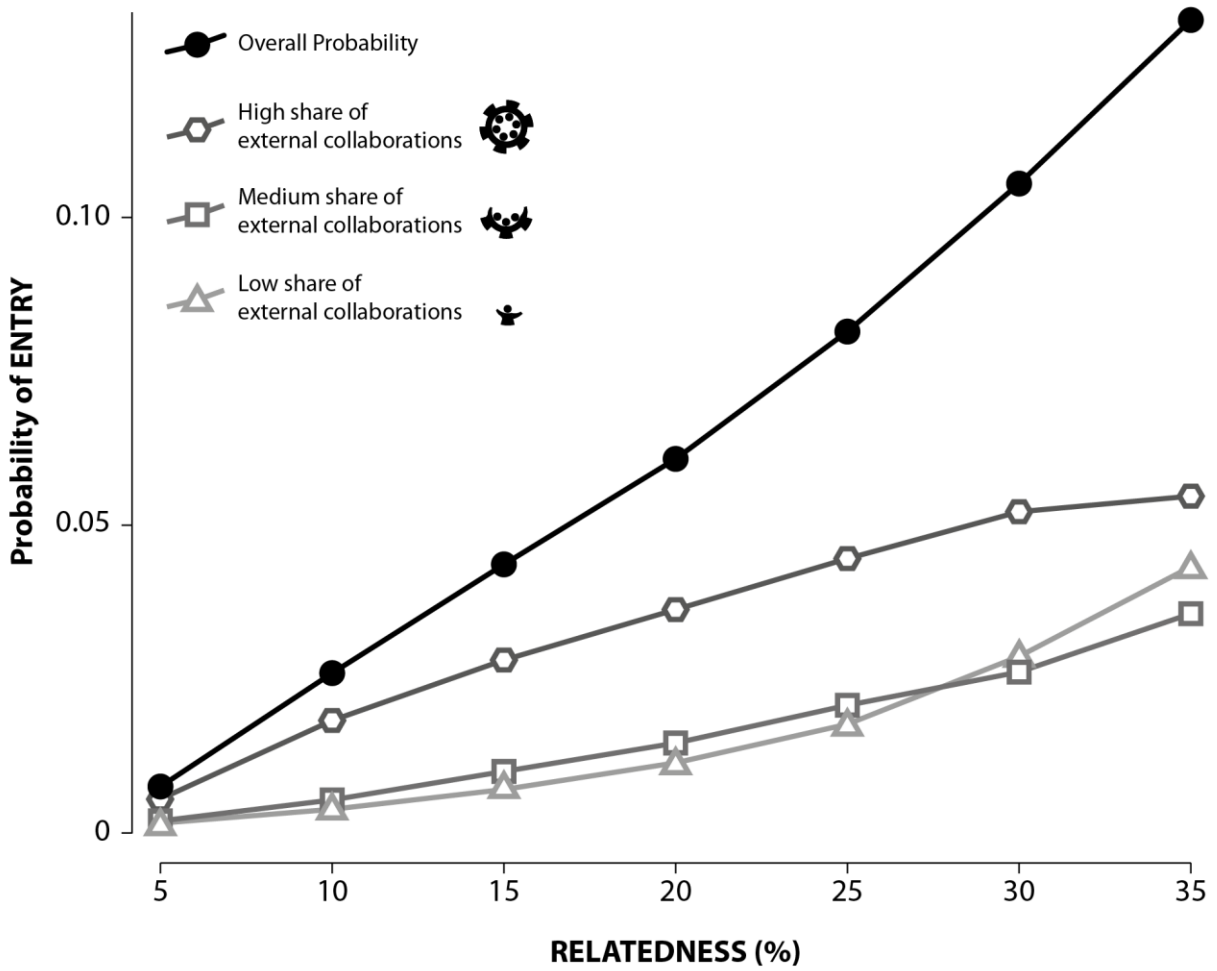
$$ENTRY_{i,r,t} = \begin{cases} 1 \; if \; RTA_{i,r,t-1} < 1 \; and \; RTA_{i,r,t} > 1 \\ 0 \; otherwise \end{cases}$$

$$RTA_{i,r,t} = \frac{patents_{r,t} \, \frac{i}{\Sigma_i \, patents_{r,t}} \, (i)}{\Sigma_r \, patents_{r,t} \, \frac{i}{\Sigma_r \Sigma_i \, patents_{r,t}} \, (i)}$$

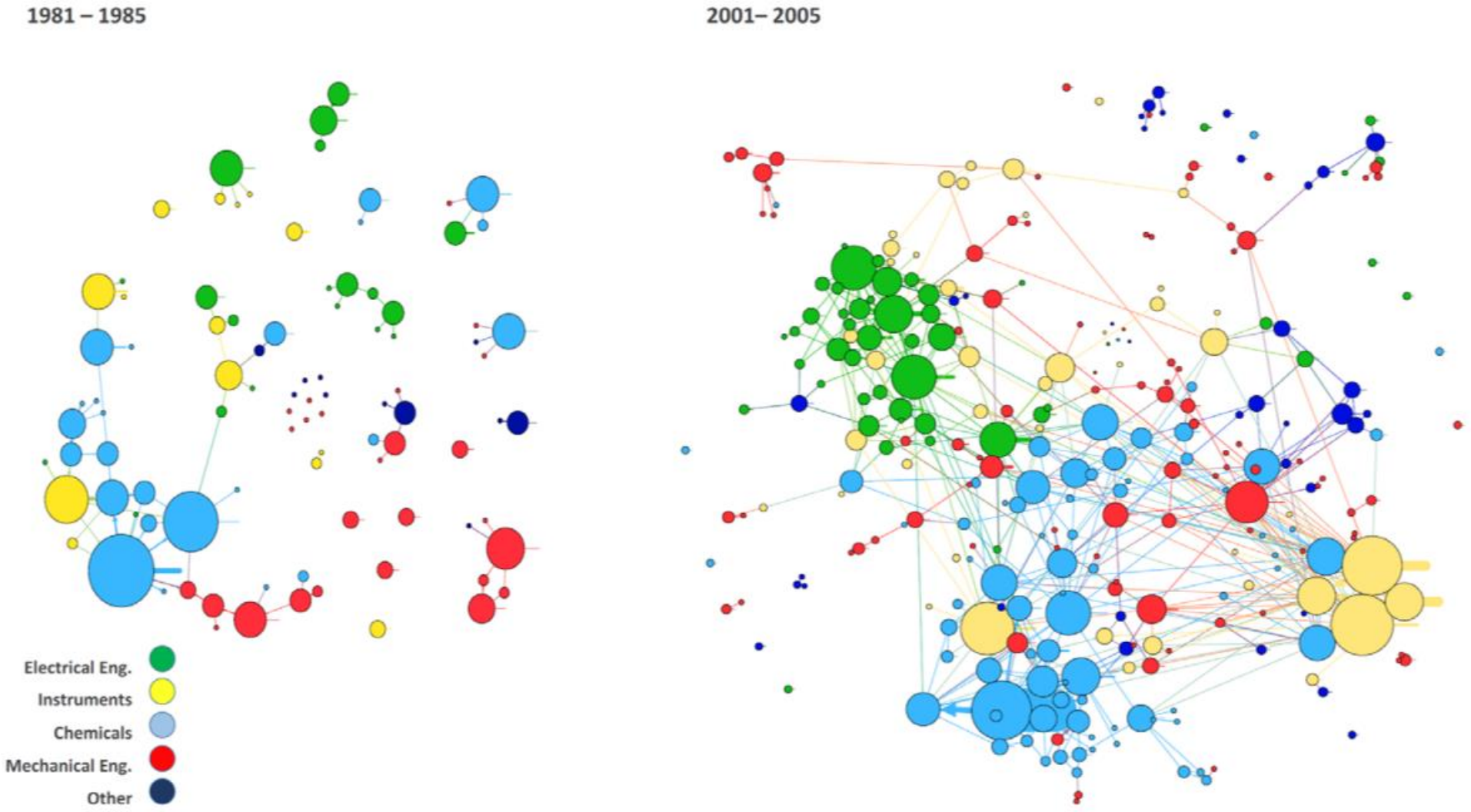
$$RELATEDNESS_{i,r,t} = \frac{\sum_{j \in r} x_j \, \varphi_{ij,t}}{\sum_j \quad \varphi_{ij,t}} \times 100$$

$$\varphi_{ij,t} = min\{P, (R|TAx_{i,t} RTAx_{j,t}) \, \{P(R|TAx_{j,t} RTAx_{i,t})\}\}$$

$$EXTCOLLAB_{i,r,t} = \frac{EL_{i,r,t}}{EL_{i,r,t} + IL_{i,r,t}}$$



The knowledge space



Kogler and Whittle (2018) Edward Elgar Handbook on Geographies of Regions and Territories

Regression analysis

Regress the probability of ENTRY on NETWORK variables and controllers.

- Region-technology fixed effects (time fixed effects).
- LPM fit (logit).
- Lagged independent variables.

$$P\left(ENTRY_{i,r,t} = 1\right) = \alpha + \beta X_{i,r,t-1} + \gamma Z_{i,r,t-1} + \mu_{i,r,t} + \varepsilon_{i,r,t}$$

$$DIVERSITY_{i,r,t} = \frac{-\sum l_{rq} \times log\left(l_{rq}\right)}{log(k)}$$

$$INTENSITY_{i,r,t} = \frac{\sum_{q} \quad L_{f \in i, r = f \in q}}{\sum_{q, f \in i, r = f \in q} \quad N_{f \in i,r} \times N_{f \in q}}$$

Explanatory and control variables

Variable Name	Description
RELATEDNESS	Relatedness density (Hidalgo et al. 2007) calculated on the knowledge-space network. Measure of how close an emerging technology *i* is to other existing technologies in the region at time *t*.
EMPLOYMENT	Total number of employees in a region.
FIRMS	The number of firms in a region and CPC class that have been assigned at least one patent until *t*.
NEWFIRMS	The number of firms in a region and CPC class that have been assigned a first patent since *t*-1.
MULTILOC	The share of firms in the region and CPC class that employ inventors in other regions as well.
INVPERFIRM	The number of inventors per number of firms that are assignees of patents in the region and CPC class at period *t*.
EXTCOLLAB	The ratio of interregional collaborations among all collaborations of the inventors living in the region and patenting in a CPC class in period *t*.
DENSITY	The ratio of observed co-inventor collaboration among all possible collaborations the inventors in a specific CPC class could have within the region.
DIVERSITY	Entropy of the aggregated co-inventor connections from a CPC class and the region to other regions at period *t*.
INTENSITY	The ratio of observed links within the firm to other regions among all possible such links. This measure is calculated at the CPC and region level at period *t*.

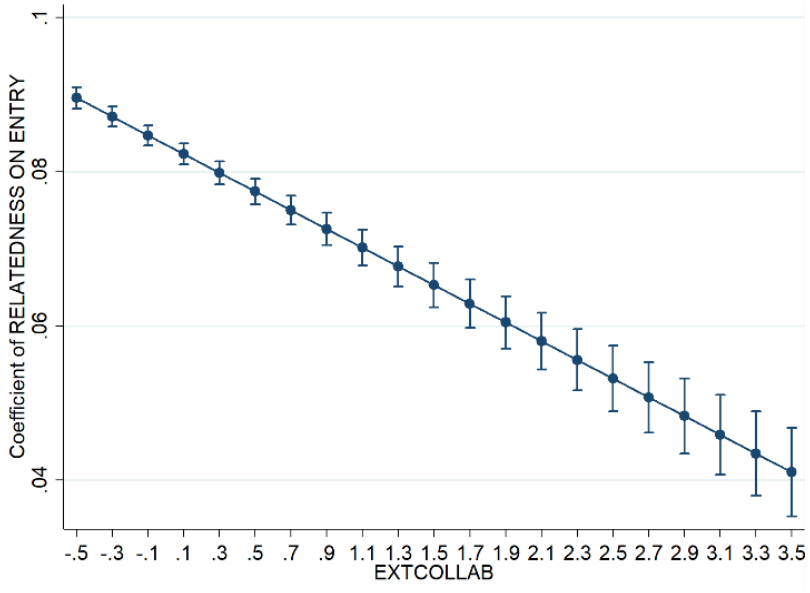
	(1) LPM	(2) LPM	(3) LPM	(4) LPM	(5) LPM	(6) LPM	(7) LPM	(8) LPM	(9) LOGIT	(10) LOGIT
	ALL REGIONS					METRO	NONMETRO	ALL REGIONS		
RELATEDNESS	0.084***	0.084***	0.087***	0.084***	0.087***	0.084***	0.089***	0.072***	0.726***	0.541***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.006)	(0.010)
EMPLOYMENT	0.025***	0.025***	0.024***	0.025***	0.024***	0.033***	0.016***	0.000	0.834***	0.350***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.022)	(0.021)
INVPERFIRM	-0.007**	-0.005*	-0.003	-0.007**	-0.005*	-0.006*	-0.003	-0.003	-0.004	0.015
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.002)	(0.009)	(0.009)
FIRMS	-0.023***	-0.023***	-0.013***	-0.024***	-0.014***	-0.012***	-0.016***	-0.009***	-0.082***	-0.054***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.002)	(0.011)	(0.011)
NEWFIRMS	0.014***	0.014***	0.010**	0.015***	0.006	0.005	0.004	0.002	0.065***	0.015
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.007)	(0.003)	(0.014)	(0.014)
MULTILOC	0.003***	0.003***	0.003***	0.003***	0.003***	0.004***	0.003***	0.003***	0.002	-0.004
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)	(0.002)	(0.003)
DENSITY	0.003**	0.003**	0.003**	0.003**	0.003**	0.003*	0.002	0.001	-0.008*	-0.022***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.004)	(0.004)

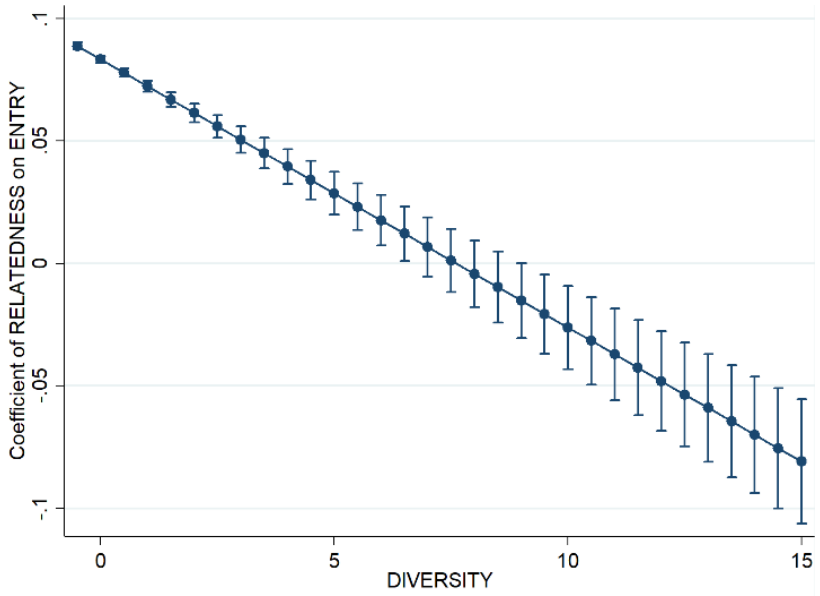
	(1) LPM	(2) LPM	(3) LPM	(4) LPM	(5) LPM	(6) LPM	(7) LPM	(8) LPM	(9) LOGIT	(10) LOGIT
	ALL REGIONS					METRO	NONMETRO	ALL REGIONS		
RELATEDNESS	0.084***	0.084***	0.087***	0.084***	0.087***	0.084***	0.089***	0.072***	0.726***	0.541***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.006)	(0.010)
EXTCOLLAB		0.002*	0.030***							
		(0.001)	(0.002)							
RELATEDNESS × EXTCOLLAB			-0.029***							
			(0.002)							
DIVERSITY				0.001	0.021***	0.023***	0.002	0.018***	0.153***	0.038**
				(0.001)	(0.003)	(0.004)	(0.007)	(0.003)	(0.014)	(0.014)
INTENSITY				0.001	0.023***	0.030***	0.021***	0.005	0.208***	0.017
				(0.001)	(0.003)	(0.004)	(0.005)	(0.003)	(0.012)	(0.012)
RELATEDNESS × DIVERSITY					-0.018***	-0.019***	0.005	-0.009**	-0.178***	-0.072***
					(0.003)	(0.003)	(0.008)	(0.003)	(0.013)	(0.012)
RELATEDNESS × INTENSITY					-0.023***	-0.027***	-0.024***	-0.006*	-0.230***	-0.049***
					(0.002)	(0.003)	(0.004)	(0.003)	(0.011)	(0.011)
CONSTANT	0.134***	0.135***	0.136***	0.134***	0.136***	0.122***	0.135***	0.228***		
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.009)		
Period FE	NO	NO	NO	NO	NO	NO	NO	YES	NO	YES
N	2,418,740	2,418,740	2,418,740	2,418,740	2,418,740	613,604	1,805,136	2,418,740	762,490	762,490
R-sq	0.017	0.017	0.017	0.017	0.017	0.016	0.018	0.022		
AIC	150,348.7	150,329.5	149,423.6	150,348.5	149,314.0	254,731.8	-169,679.4	138,328.0	520,371.7	511,581.8
ll	-75,167.3	-75,156.8	-74,702.8	-75,165.2	-74,659.5	-127,354.9	84,850.7	-69,148.0	-260,174.9	-255,774.9

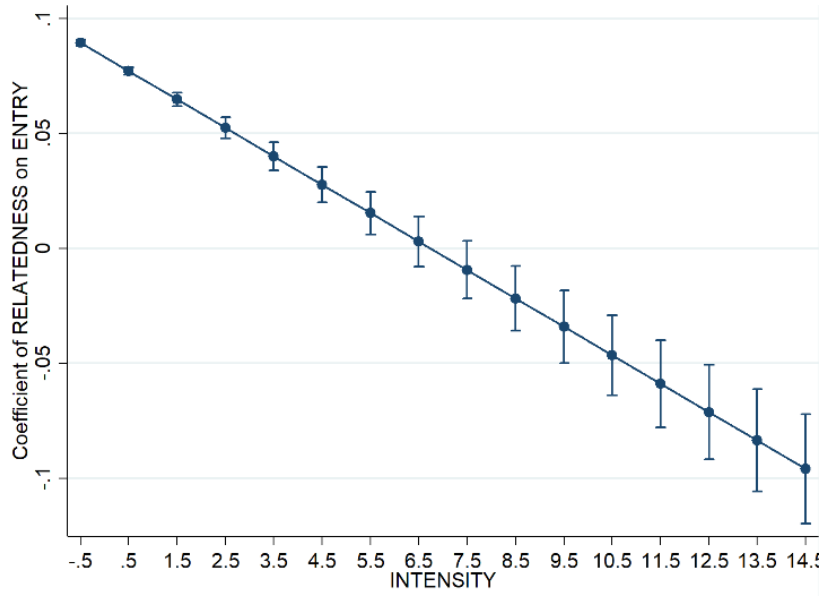
Interregional collaboration as substitution for relatedness?

Coefficients of RELATEDNESS at levels of EXTCOLLAB, DIVERSITY, INTENSITY.

- EXTCOLLAB: RELATEDNESS becomes smaller but remains positive.
- DIVERSITY, INTENSITY: RELATEDNESS becomes negative at their high levels.



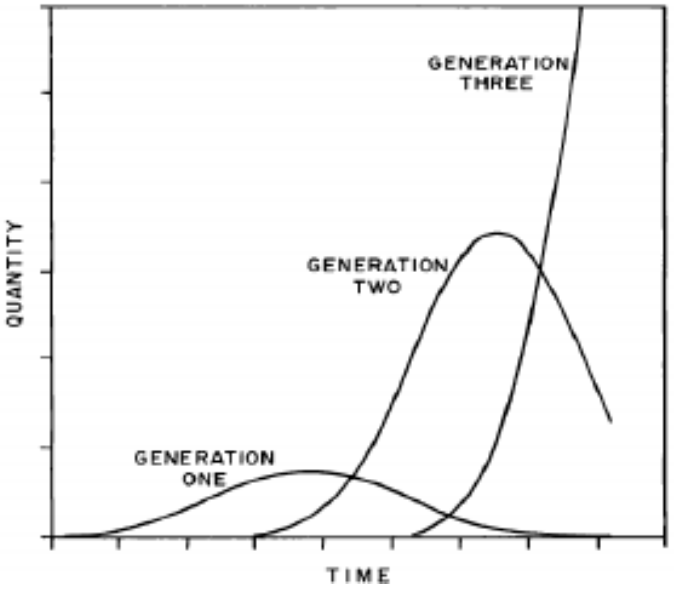




3. Spatial life cycles – extension of EXIT in the diversification literature

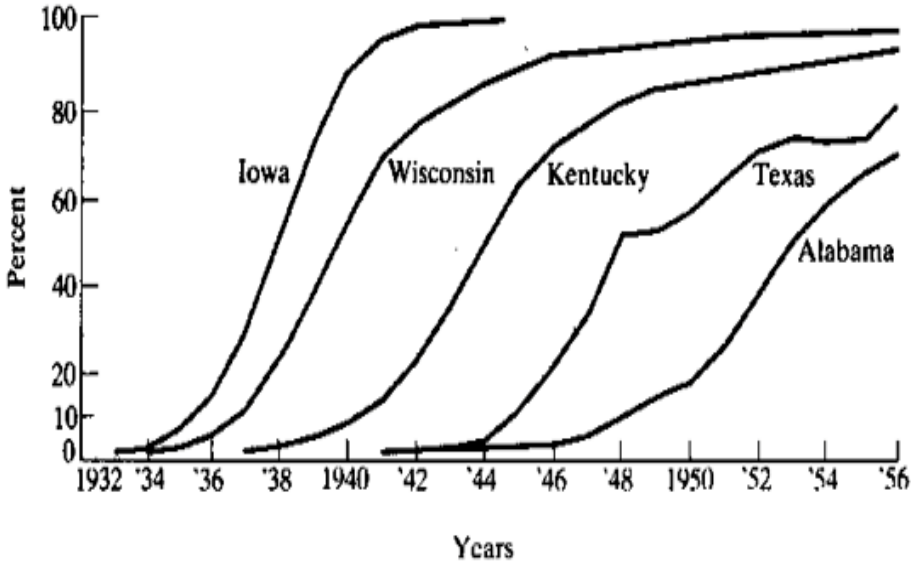
Waves of technology life-cycle

New technologies outcompete old ones.



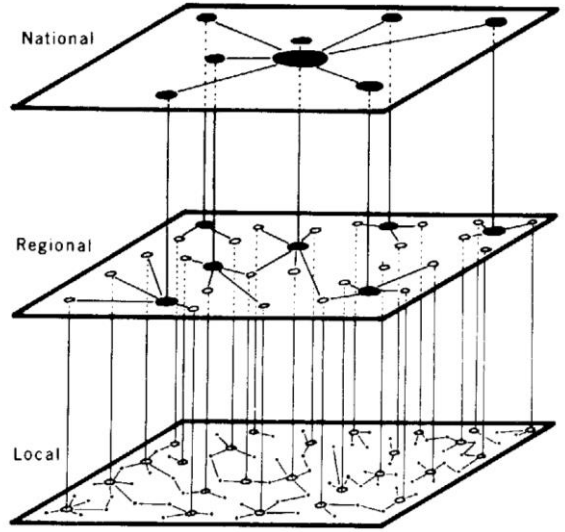
Norton & Bass (1987, Man Sci)

Innovation gets quicker to proximate places.

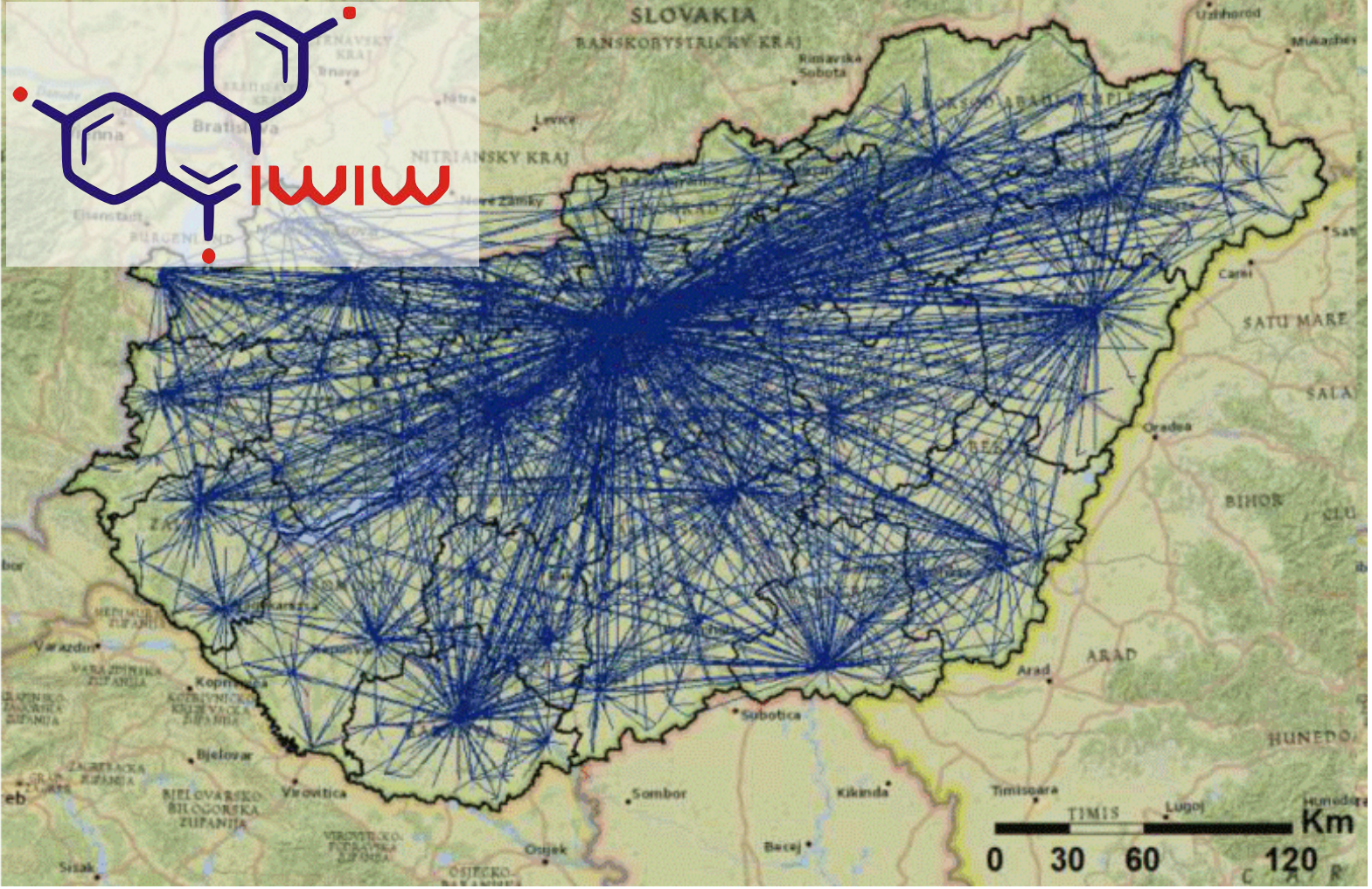


Griliches (1957, Econometrica)

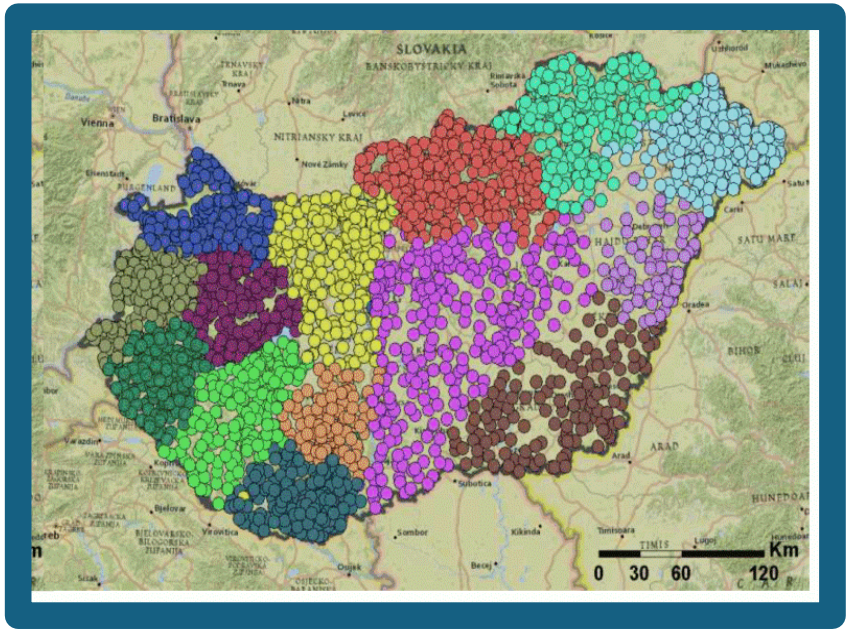
Innovation spreads from big to medium cities and then to small towns.



Haegerstrand (1953, Chicago UP





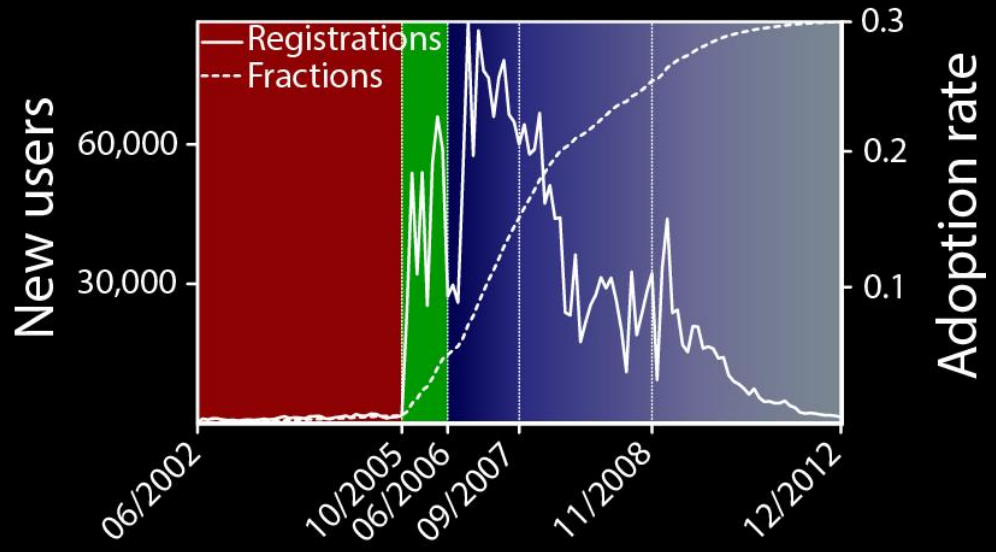


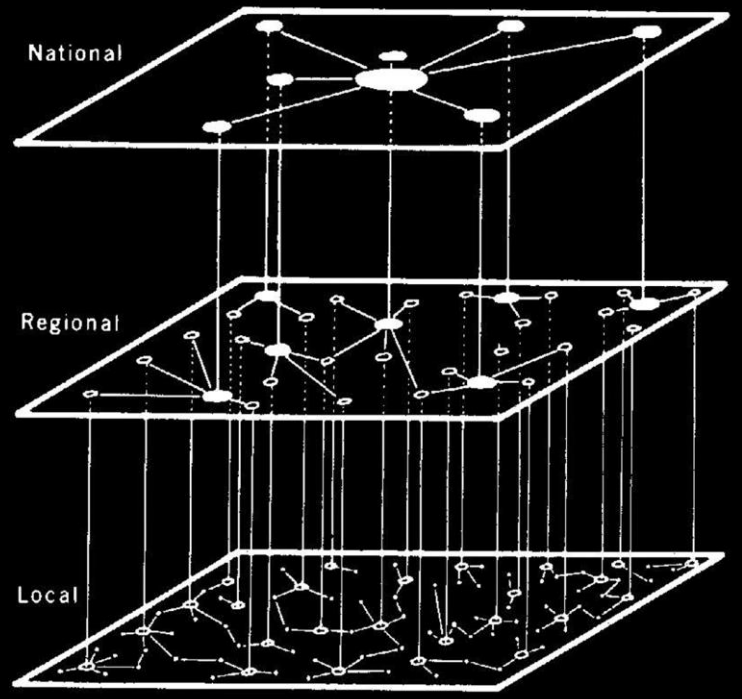
Social media platform

2.8 M Users

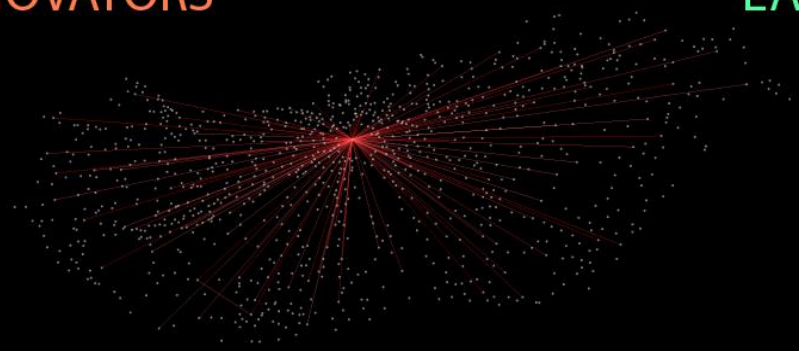
- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**

Jakobi and Lengyel (2014) Springer; Lengyel et al. (2015) PLoS ONE





INNOVATORS



EARLY ADOPTERS



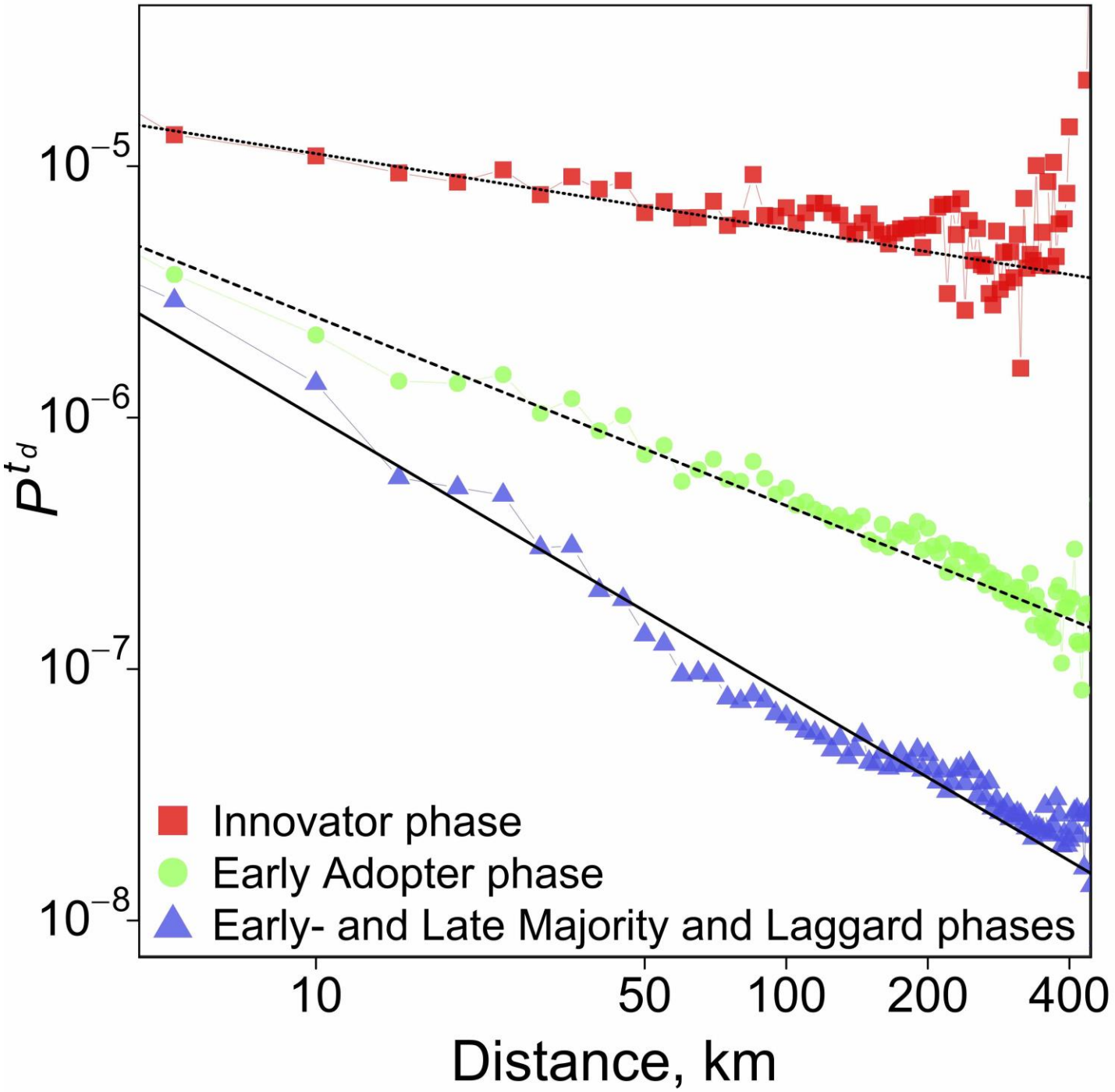
MAJORITY & LAGGARDS

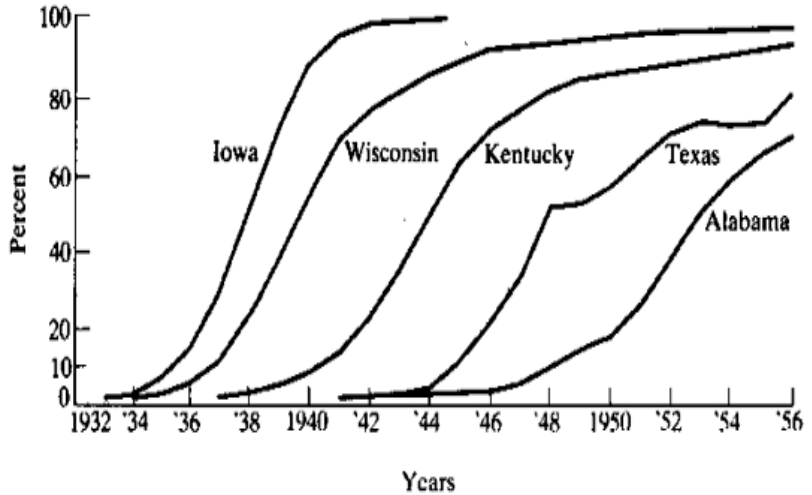


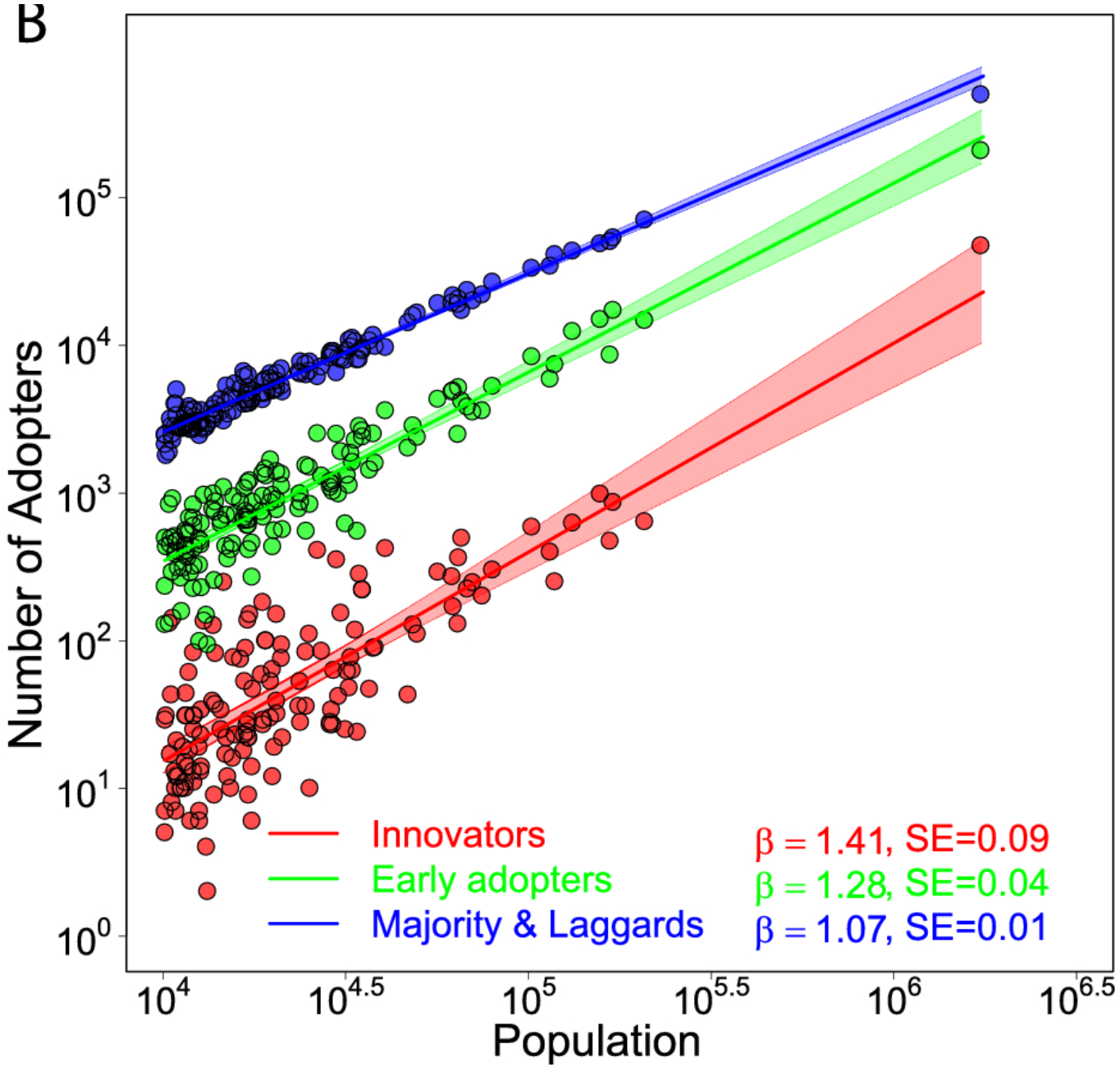
The role of distance

Distance only slightly decreases the probability of invitaiton in early phases

Diffusion becomes local in later phases.



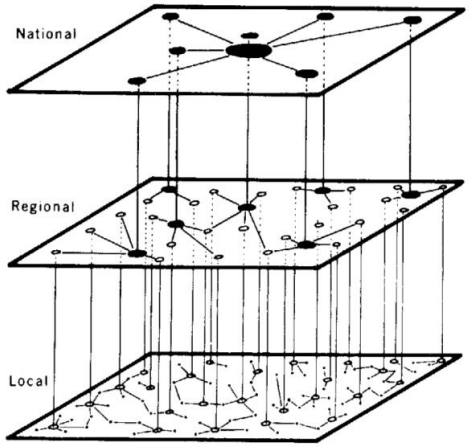




The role of size

Innovators and Early Adopters concentrate in cities.

Majority and Laggards are proportional to town size.



Spatial wave of adoption and churn



The spatial life cycle hypotheses

H1: *„The churn from an innovation is early in places that have adopted it early.”*

- Declining relative advantage and social value after saturation
- Adoption in periphery happens when centrum saturates
- Spatial differences in perceived novelty
- Value fades away

- Assumption: like adoption, churn also happens in identical spatial networks
- Mixture of diffusion of obscolescence, new waves, local network economies

The spatial life cycle hypotheses

H2: „*Not only adoption but also churn starts in large cities.*”

- Early adoption and churn are both urban phenomena.
- Cities are more innovative and more susceptible for innovation.
- Saturation is early – obscolescence is also early.

- New products can also arrive early.

The spatial life cycle hypotheses

H3: „*Churn is earlier in those locations, where the adoption of the replacement innovation cycle is early.*”

- New technologies are adopted on the basis of the old one.
- Spatial diffusion is similar across waves.
- The new replaces the old.

The spatial life cycle hypotheses

H4: „*The innovation's embeddedness in local socio-technological systems prolong its life cycle in places.*"

- Local network economies slow down churn.
- Benefits of the old technology prevail if it is embedded in local systems.

- Diffusion and churn happens in dynamic networks: local interdependencies emerge
  - Early adopters have more time to develop interdependencies
  - The networks of late adopters function more as channels of obscolescence diffusion



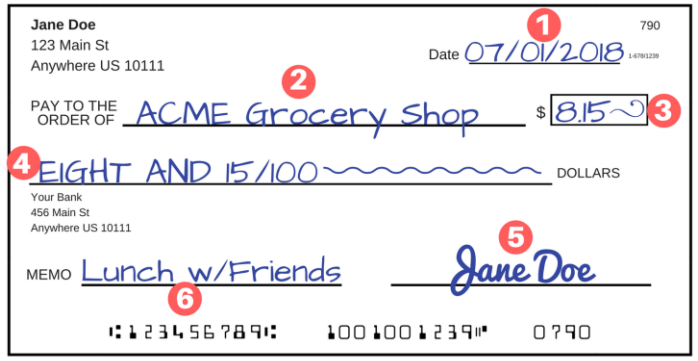
Not just social media:

(H2) Innovation is concentrated in cities (Bettencourt et al. 2007)

(H3) All IT waves come from the Bay area.

(H4) Tightly connected innovation systems tend to stick to old technologies (Grabher, 1993).

This path dependency is strong in networked products.



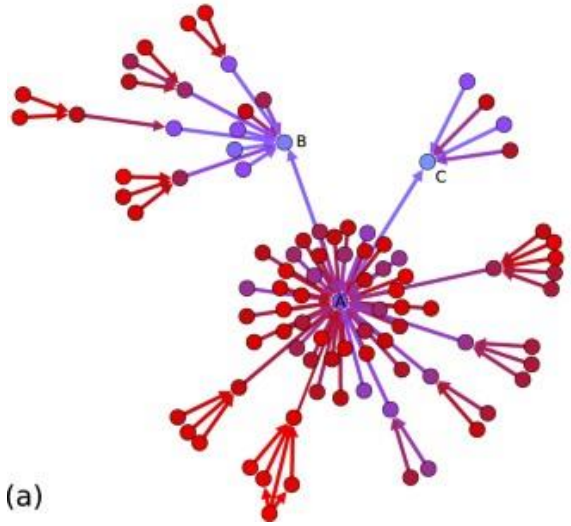
Previous research on social influence in churn

Lőrincz et al. 2019 Social Networks:

- Central individuals churn later – the network is of greater value for them.

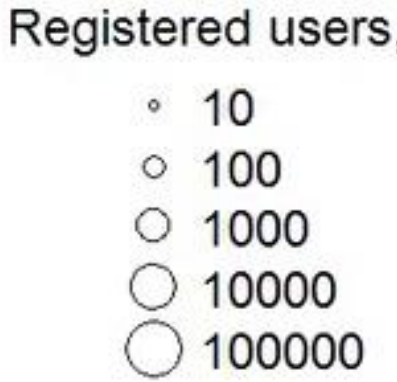
Török and Kertész 2017, SRep

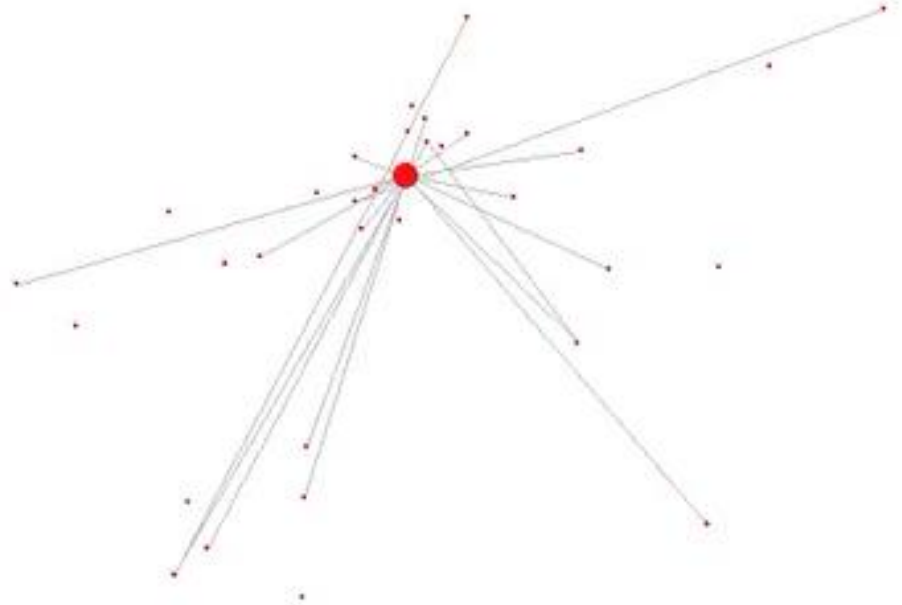
- The cascade of churn starts from the peripheral actors and their churn triggers central actors.

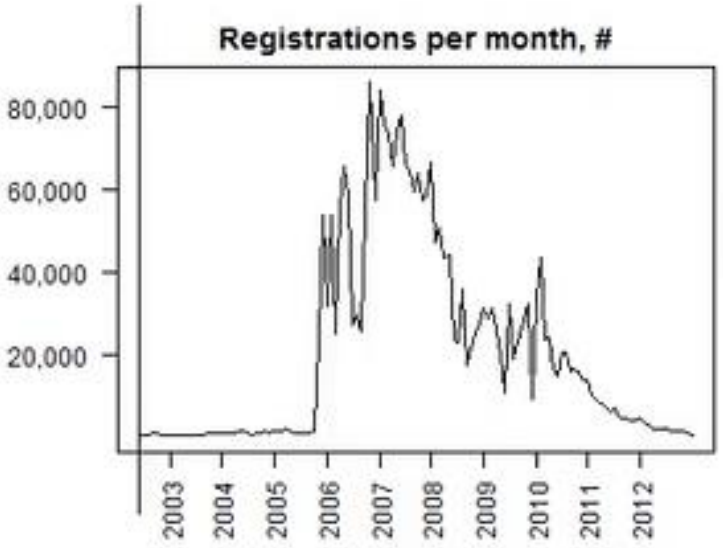


Török and Kertész (2017, Scientific Reports)

Full life-cycle of a social media platform

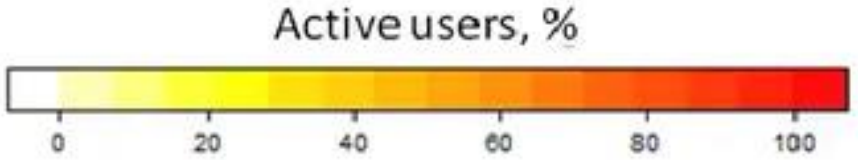




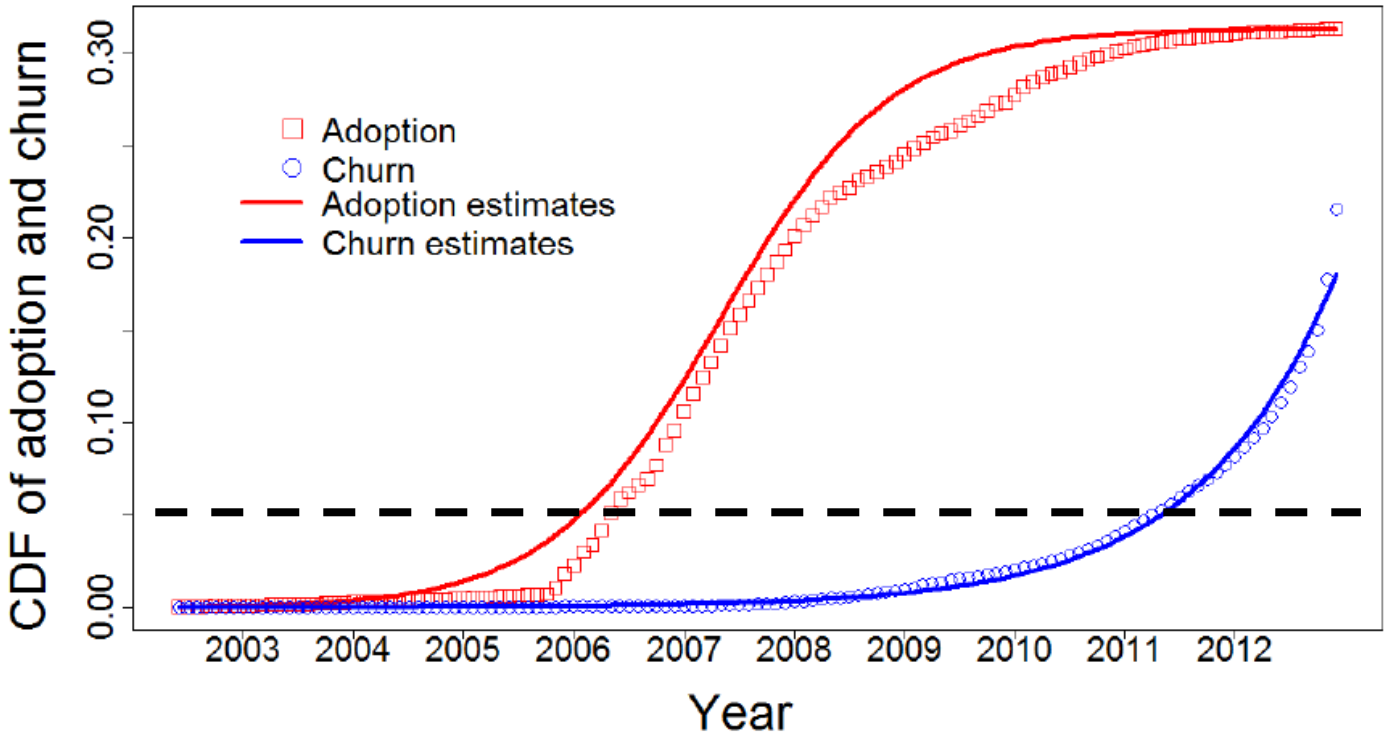


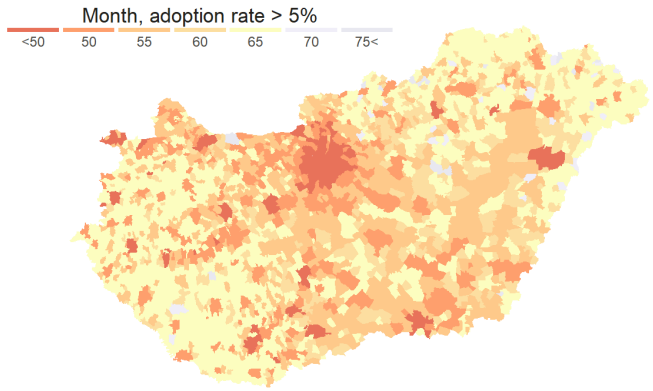
2.8 M Users

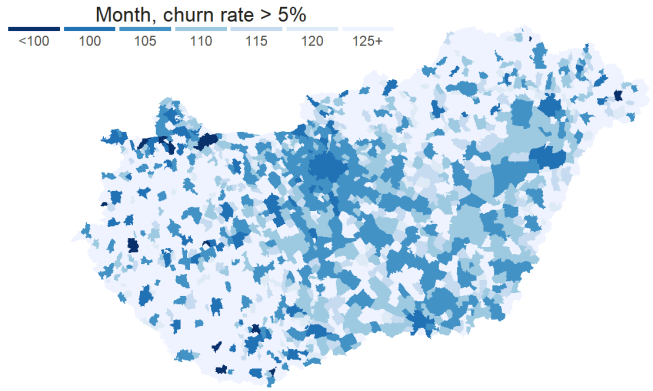
- **Location** (2557 towns),
- **Date of registration,**
- **Date of last login,**
- **Connections (time-stamp),**
- **ID of invitors**

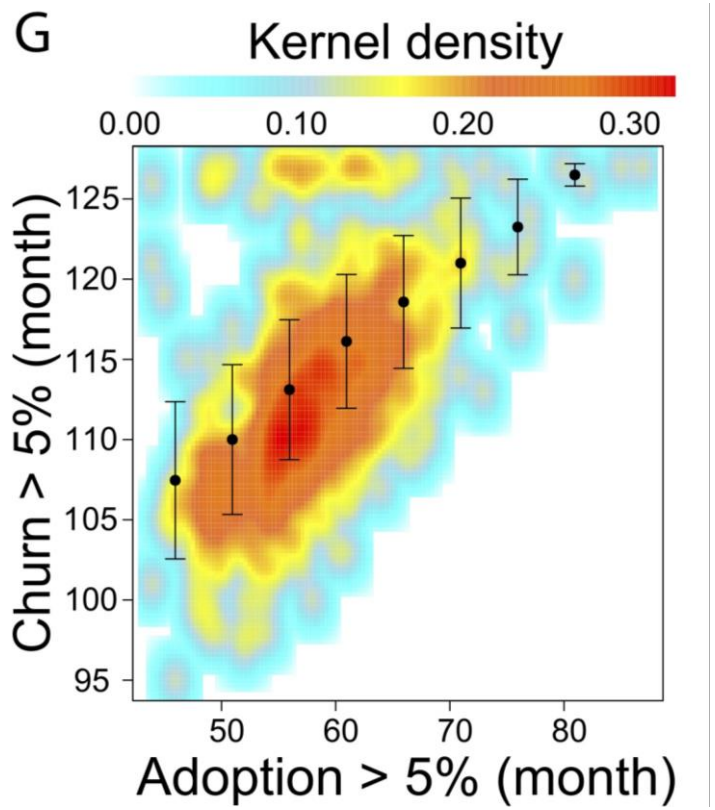


Those places churn early that adopt early

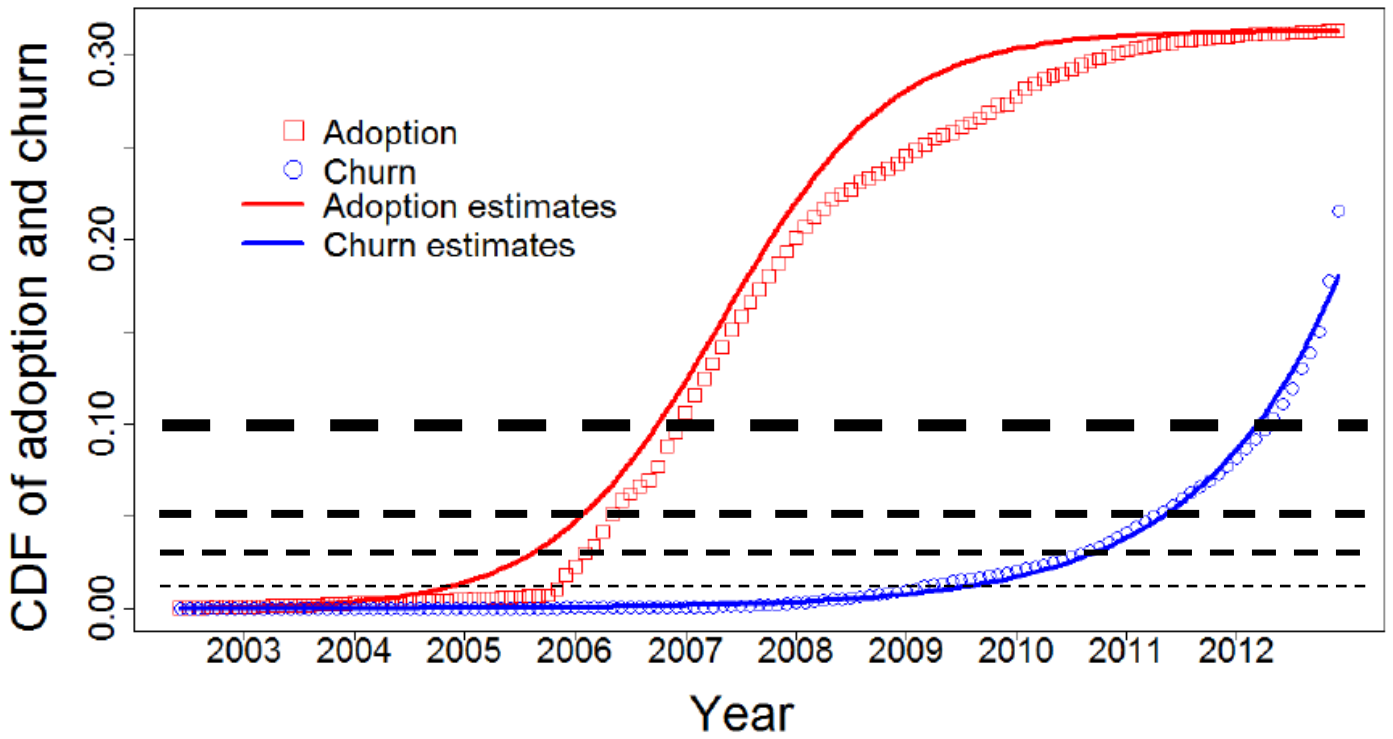


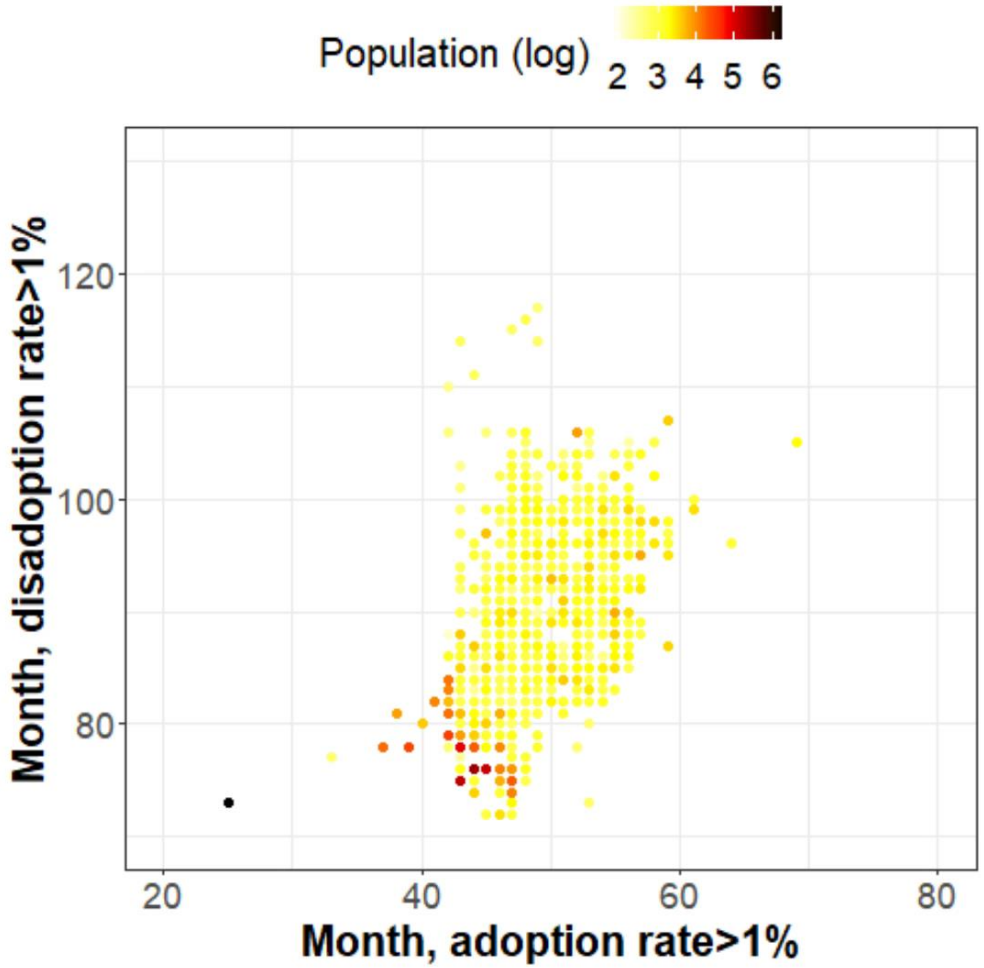






Those places churn early that adopt early





Adoption Rate

(Bass Curve)

$$\frac{dy_a(t)}{dt} = (p_a + q_a y_a(t))(1 - y_a(t))$$

Churn Rate

(Exponential growth)

$$\frac{dF_{ch}(t)}{dt} = q_{ch} F_{ch}(t)$$

of adopters

$y_a(t)$

Innovation Par.

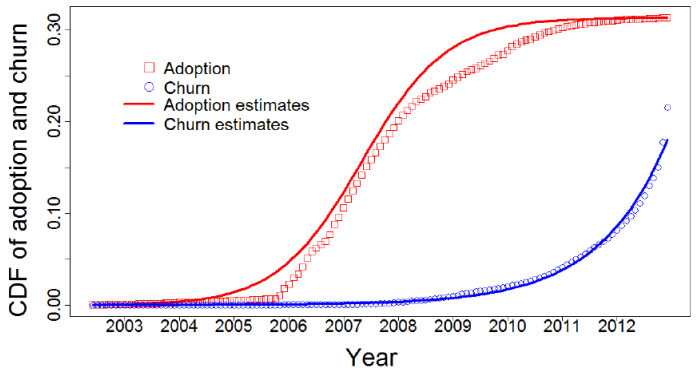
p_a

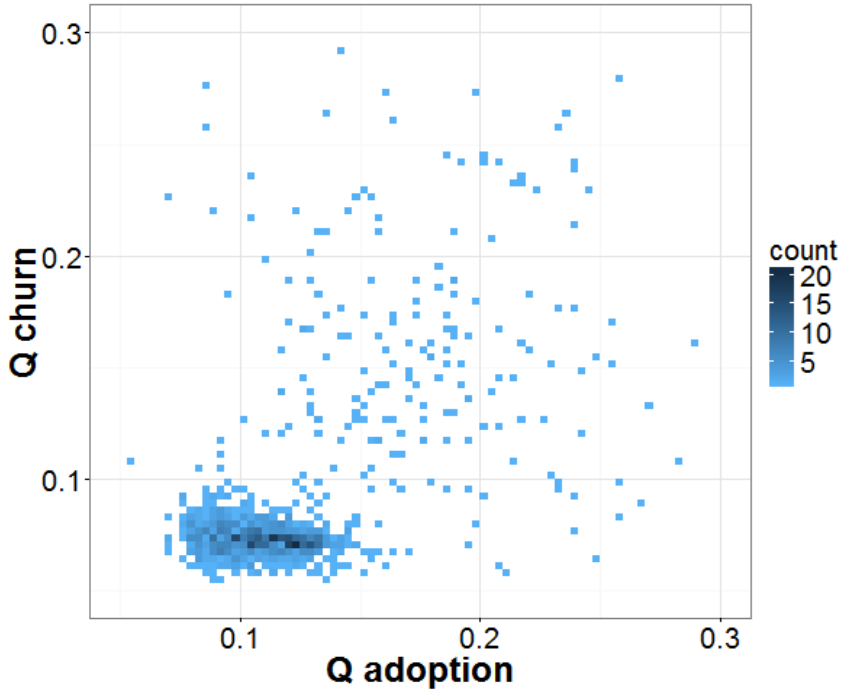
t

Time (months)

q_a

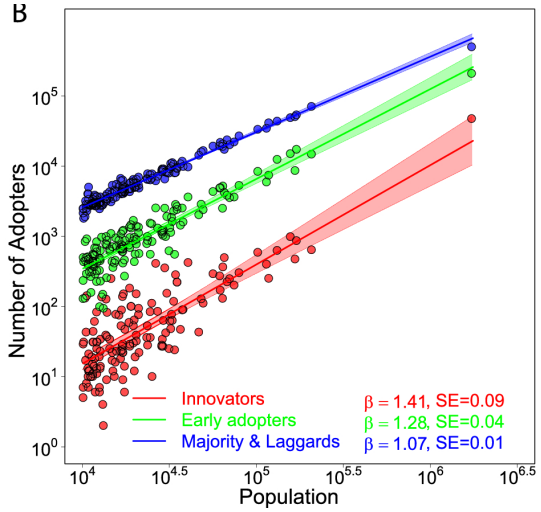
Imitation Par.

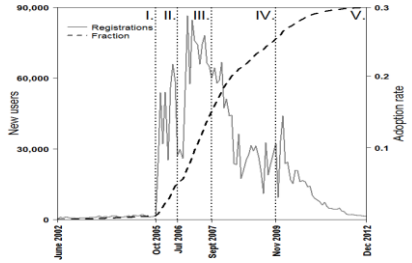


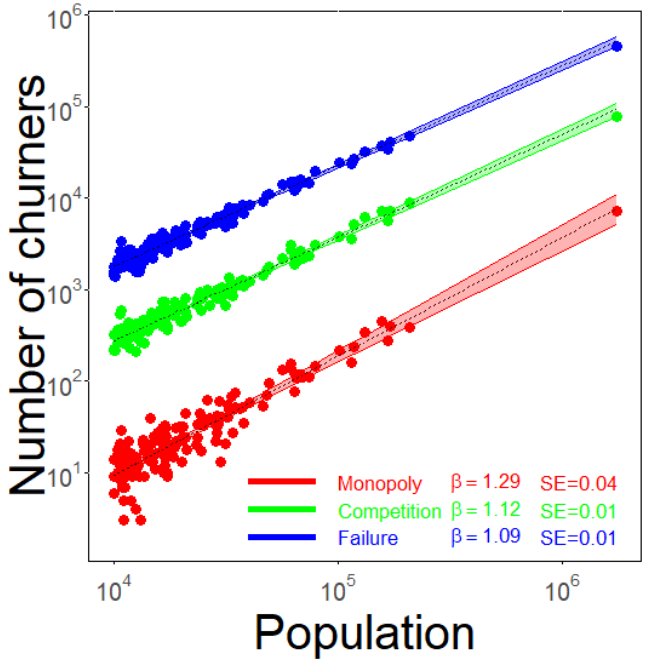


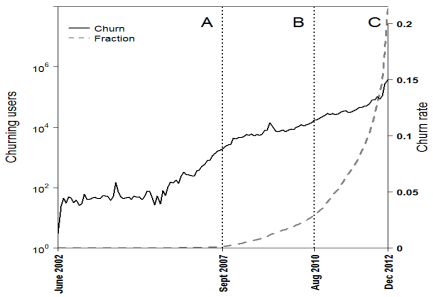
ANETILAB

The role of city size in adoption and churn



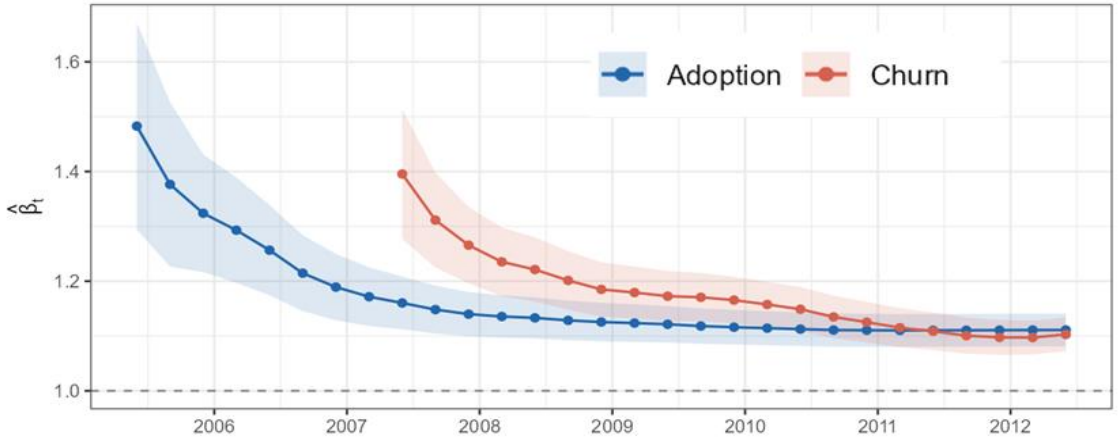




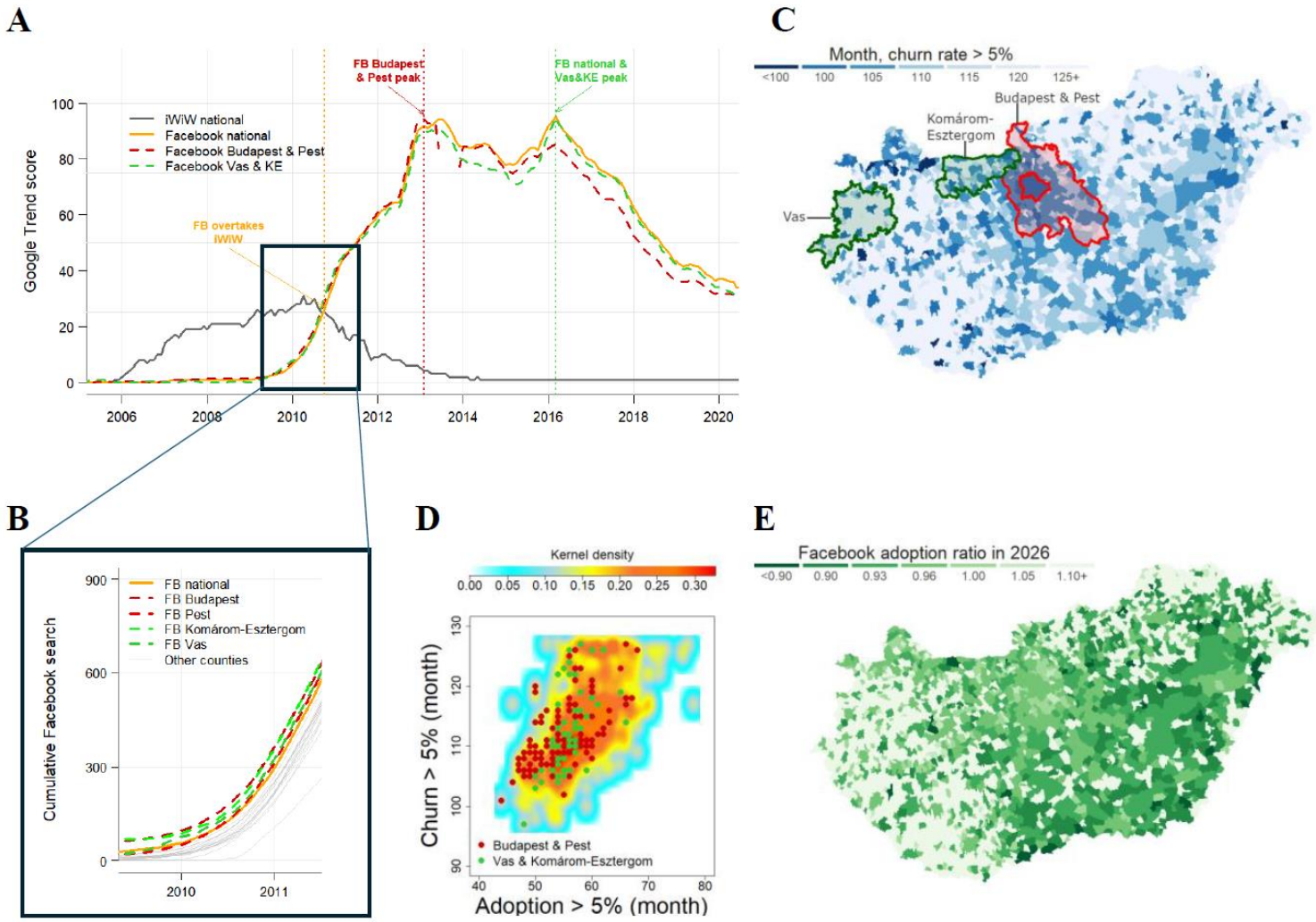


Innovators and Early Adopters concentrate in cities.

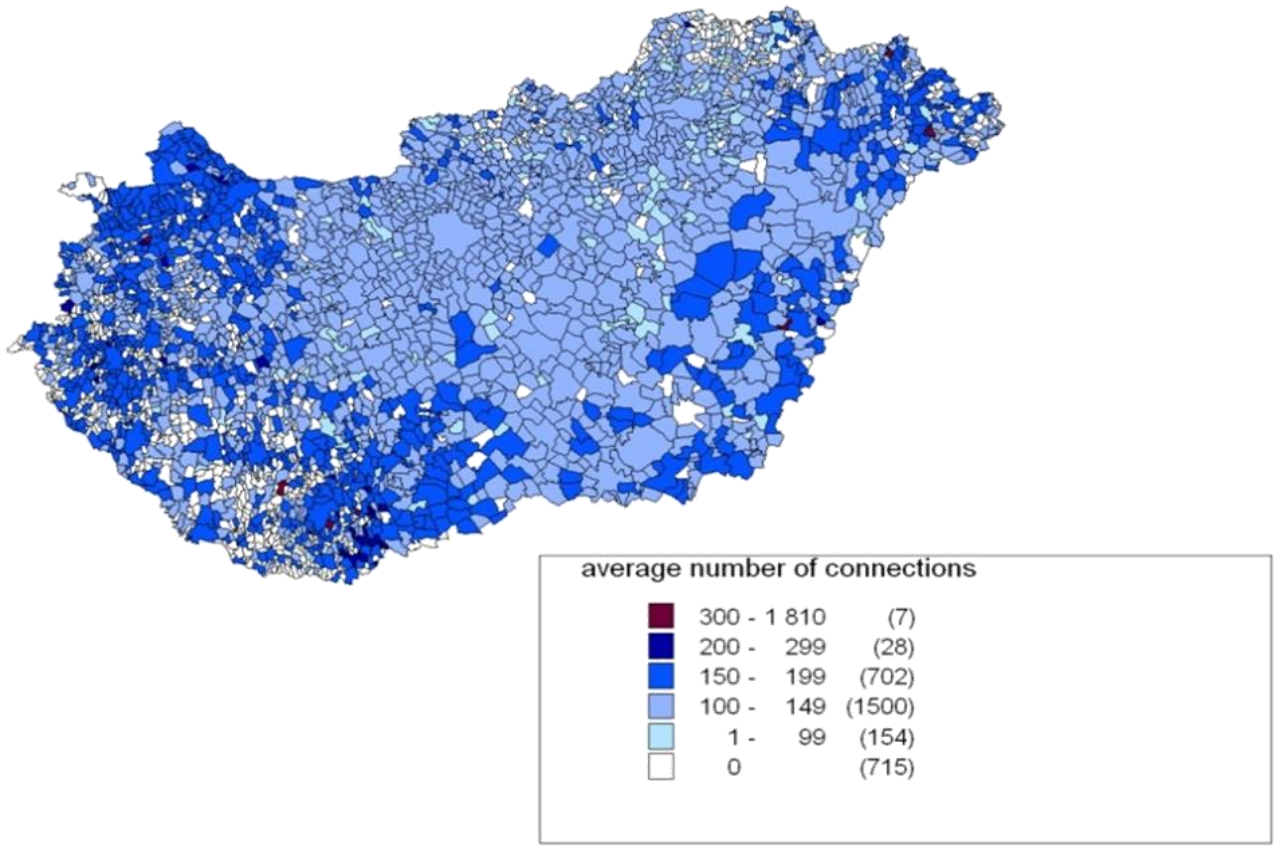
Lengyel et al. (2020, Scientific Reports)

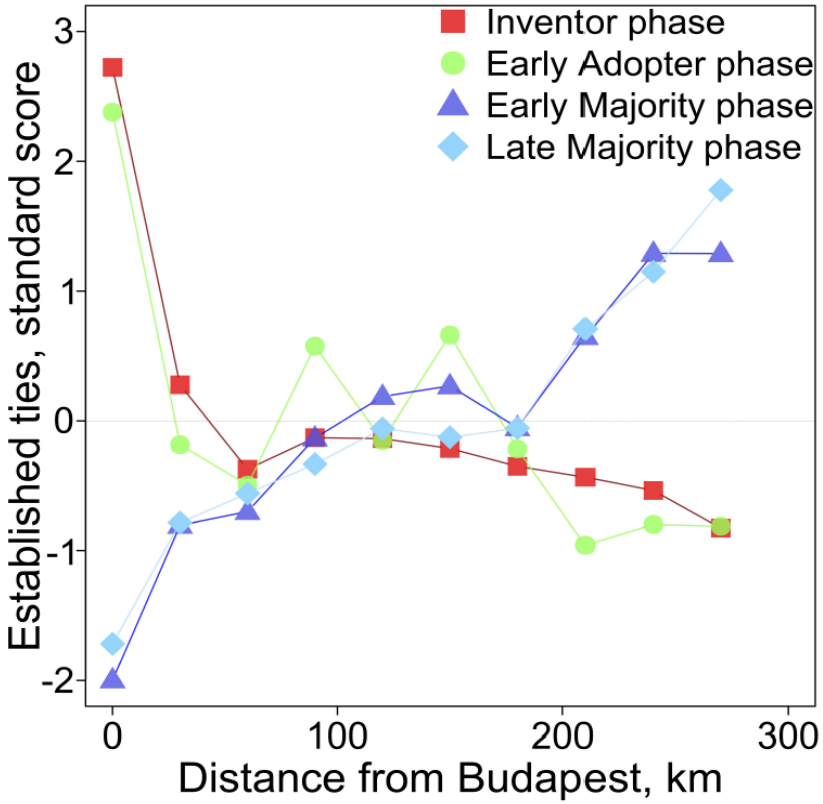


The arrival of the new technology

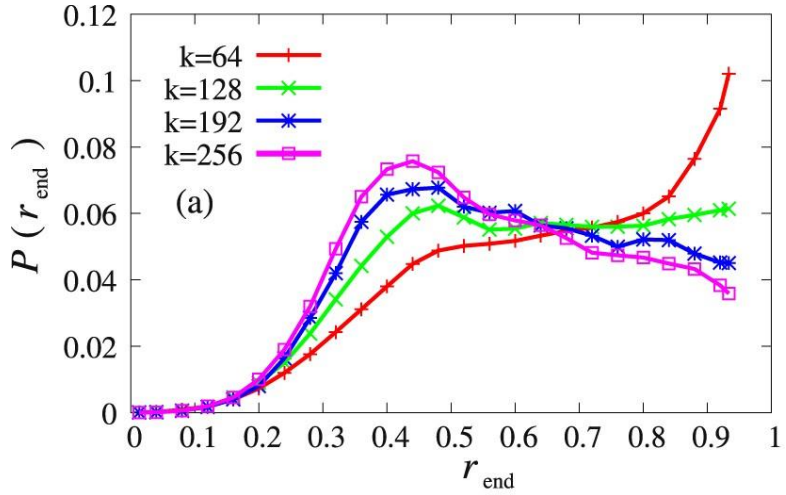


Big waves of technology usage: reporting friends

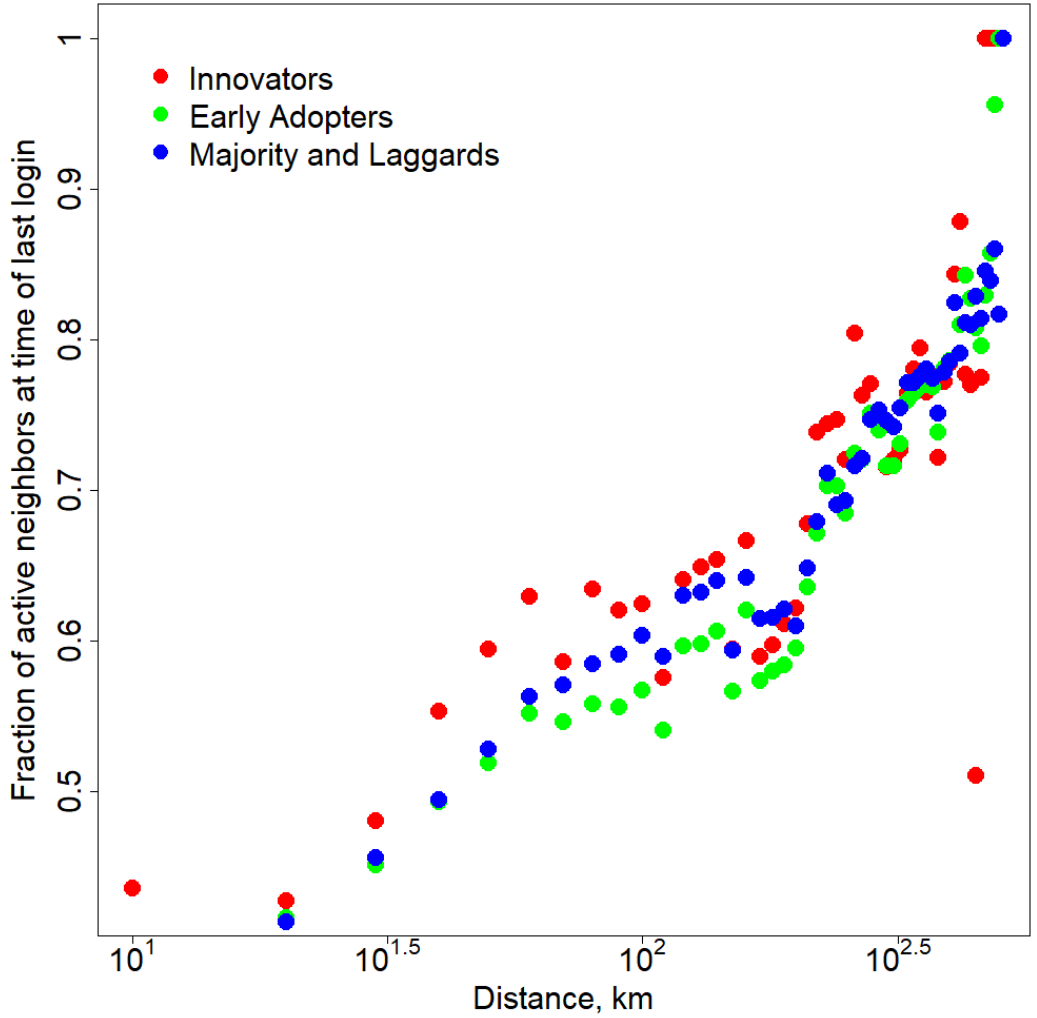




Peer effects in churn are local

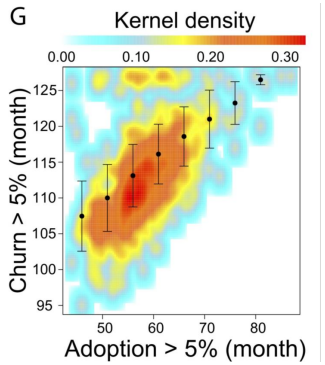


Török and Kertész (2017, Scientific Reports)



ANETILAB

Determinants of churn dynamics, town-level OLS regression. Dependent variable: Month when town-level churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations ***, *, * stand for statistical significance is reported at the 1%, 5% and 10% levels.



	Churn, 5% (month)						
	(0)	(1)	(2)	(3)	(4)	(5)	(6)
Month of adoption 5%	0.865***						0.984**
	(0.035)						(0.050
Population, log		-7.601***			-7.669***	-7.591***	-5.342**
		(0.424)			(0.521)	(0.525)	(0.458
Distance from Center, log			2.945***		-2.158***		-3.465
			(0.717)		(0.792)		(0.676
Central Region (dummy)				-2.793***		-0.391	
				(0.612)		(0.613)	
Constant	62.961***	139.956***	107.409***	113.702***	154.070***	147.757***	76.562**
	(2.033)	(1.499)	(1.446)	(0.238)	(3.579)	(2.942)	(7.056
Town Controls	NO	NO	NO	NO	YES	YES	YES
N	1,021	1,021	1,021	1,021	1,021	1,021	1,021
R2	0.377	0.240	0.016	0.020	0.302	0.297	0.496
F	617.344***	321.966***	16.873***	20.814***	73.155***	71.490***	142.385

Social connections and churn dynamics, town-level OLS regression. Dependent variable: Month when town-level churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations ***, *, * stand for statistical significance is reported at the 1%, 5% and 10% levels.

	Churn, 5% (month)		
	(1)	(2)	(3)
Month of adoption 5%	0.838***	0.933***	1.169***
	(0.037)	(0.041)	(0.054)
Population, log	-5.102***	-5.306***	-5.630***
	(0.384)	(0.412)	(0.461)
Avg. degree	-0.045***		
	(0.005)		
Avg. degree, Early Majority		0.059***	0.069***
		(0.009)	(0.009)
Avg. degree, Late Majority		-0.085***	-0.081***
		(0.008)	(0.008)
Avg. Degree, Laggards		0.000***	0.000***
		(0.000)	(0.000)
Constant	91.058***	81.273***	58.084***
	(2.831)	(3.384)	(5.401)
Town Controls	NO	NO	YES
N	1,021	1,021	1,021
R2	0.492	0.512	0.535
F	328.188***	213.014***	129.173***

Further steps: generalization and synergies

Different contexts

- Web technologies
- Technologies from patent (and trademark) data
- Scientific terms, concepts, keywords in publications
- Topics and expressions in news

Mechanisms

- Better identification of substituting technologies

Theory and models

- Recovery like in the network epidemiology models
- Regional economics and economic development